

West Bengal State Electricity Distribution Company Limited

(A Govt. Of West Bengal Enterprise)



Technical Specification for SCADA compatible indoor 33 kV extensible, composite GIS switchgear cum control, relay and metering panel

September'2023

Technical Specification for 33 kV SCADA compatible indoor GIS switchgear panel

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Technical Specification for SCADA compatible indoor 33 kV extension composite GIS switchgear cum control, relay and metering panels.

1. GENERAL REQUIREMENT:

GIS Switchgear shall be an indoor gas insulated and metal-clad cubicle design with single busbar system in accordance with single line diagram and data sheet. Each panel shall be metal enclosed, free standing, fully arc proof, floor mounting, flush fronted and arranged to single structure with a common busbar assembly. Each compartment shall be protected by a metal enclosure with IP 65 minimum for gas compartments and IP4X for the supporting frames, low voltage and other compartments, IP3X for cable compartment and IP2X drives compartment, including cable entry, shall be vermin proof.

Adequate safety margin with respect to thermal, mechanical, dielectric stress, dynamic short circuit fault and insulation coordination be maintained during design, selection of raw material, manufacturing process etc. so that the GIS provides long life with least maintenance.

The complete switchgear shall be designed to manage the risks associated with it such that there shall not be any safety hazard to the employees in normal service and during inspection and maintenance. The design should be such that all parts subjected to wear and tear are easily accessible for maintenance purposes. The Service Class Continuity of Switchgears shall be LSC 2 (as per IEC 622771-200).

The manufacturer shall give guarantee for maximum leakage rate of SF₆ gas will be lower than 0.1 % per year. SF₆ gas leakage rate should not exceed 0.1% per annum after guarantee period. In case of Gas Leak the GIS should have the capability to withstand dielectric strength at minimum 1 bar pressure. Separate gas monitoring sensors should be available for all the gas filled chambers.

Gas Pressure relief device/Explosion Vent/Pressure Relief duct shall be provided for each SF₆ gas compartment, so that in case of a fault in a compartment, the gases produced are safely vented out, thereby minimizing the possibility of its spreading to other compartments and panels.

The panel complying ingress protection IP 65 for HV live part and IP 4X for LV compartment shall be constructed from corrosion-resistant stainless steel sheet of min 2 mm thickness or high grade Aluminum alloy with min 6mm thickness, filled with SF₆ accommodating the primary switching devices (VCB and Three position disconnecter cum earthing switch) and all live parts.

The panels must be suitably treated and powder coated with minimum 60 micron thickness, to achieve indoor worthiness and corrosion protection and should pass salt spray test for not less than 500 hrs.

2. SCOPE OF SUPPLY:

This scope of work shall include design, engineering, manufacture, factory production & shop testing, inspection, packing, dispatch, loading, unloading at site and field service assistance during installation and commissioning of SF₆ gas-insulated vacuum circuit breaker switchgear and associated equipment of complete 33kV GIS Switchgear.

The scope includes all material, equipment and works required for the erection of the panels with all items considered essential for safe and trouble-free continuous commercial operation of the system in a manner acceptable to the Owner and complying with latest revision of national and international standards Codes & Practices, Indian Electricity Rules and Indian Electricity Act.

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3. SERVICE CONDITIONS:

All materials supplied and installed shall be capable of operating without fault in a tropical climate, which exhibits a high level of ultra-violet radiation and severe thunderstorms. Relevant environmental conditions are listed as follows:

1	Max. ambient air temperature	50°C
2	Min. ambient air temperature	(-) 5°C
3	Average Daily Max. ambient temperature (Design Ambient)	40°C
4	Max. yearly weighted average ambient temperature	32°C
5	Max. altitude above mean sea level	1000m
6	Relative Humidity(%)	10 - 100
7.	Avg. No. of Thunder storm Days / Year	100
8	Avg. annual rainfall	3000 mm
9	Maximum wind pressure	250 kg / sq. mtr.
10	Pollution Level	Heavily polluted area
11	Seismic Level	Zone level III & IV

4. System particulars:

Parameters	HV Side	LV Side
Nominal system voltage	33 kV	11 kV
Corresponding highest system voltage	36 kV	12 kV
Frequency	50 Hz±3%	50 Hz±3%
Number of phases	3	
Neutral earthing	Grounded through Earthing Transformer	solidly earthed
Impulse frequency withstand voltage (kVp)	170	75
Power frequency withstand voltage (kV rms)	70	28
Power cable size	3- core 300 / 400 sq mm or 1-core 400/630/1000 sq mm	

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**5. REFERENCE STANDARDS:**

Standard's Description	Standard Name / No
Relevant safety regulation of CEA	Indian Electricity Rules 1956
Latest edition	Indian electricity act 1910
IEC 62271-200	AC metal-enclosed switchgear & control gear for rated voltage above 1 KV and up to and including 52 KV
IEC 62271-1	High Voltage switchgear and controlgear-Part-1: Common specification
IEC 62271-100	High voltage Alternating current circuit breakers.
IEC 60694	Common clauses for high voltage switchgear and control gear standards
IEC 62271-103	High voltage switches for rated voltages above 1 KV and less than 52 KV.
IEC 60071	Insulation coordination
IEC 61128	Alternating current disconnectors bus transfer current switching
IEC 61129	Alternating current earthing switches-induced currentswitching
IEC 60265	High voltage switches
IEC 60137	Bushings for alternating voltages above 1000 V
IEC 60233	Tests for hollow insulators for use in electrical equipment
IEC 61259	Requirements for switching of bus charging current by GIS disconnectors
IEC 60059	IEC Standard current Rating
IEC 62271-102	A.C. disconnectors (isolators) and earthing switchesfor voltages above 1000 V
IEC 62271-207	Seismic qualification for gas –insulated switchgears assemblies for rated voltages above 52 KV
IEC 60376	New Sulphar Hexafluoride (SF ₆)
IEC 60480	Guidelines for checking and treatment of Sulphar Hexafluoride (SF ₆)
IEC 61633	High voltage Alternating Current Circuit Breakers-Guide for Short Circuit andSwitching Test-Procedure for Metal Enclosed and Dead tank Circuit Breaker.

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IEC 61634	High voltage Switchgear and Control gear use and handling of SF ₆ in Highvoltage Switchgear and Control gear
IEC 61243-5	Voltage detection systems
IEC 60044-1, IS 16227, IS 2705	Current Transformers
IEC 60044-2, IS 16227, IS 3156	Potential transformers
IEC 62271-209	Cable connections for gas insulated switch gears
IS 5621:1980	Hollow Insulators for use in electrical equipment
IS:2544/1973	Porcelain Post Insulators
IS 8828/1996	MCB
IS 12063/1987	Degree of protection provided for enclosures for electrical equipment.
IS 5/2005	Colors for ready mixed paints and enamels.
IS 55781984	Marking of insulated conductor.
IS 1248/2003	Indicating instruments.
IS 6875 amended up to date	Control switches.
IS 3231/1986 & 87 amended up to date	Electrical Relays for Power System Protection.
IEC 60255 amended up to date	Numerical protection relays.
IS 8686/1977	Static protective relays.
IS 4794/68 & 86	Push button.
IS:9385/1979	High Voltage Fuses
IS 9431/1979	Indoor post insulator of organic material
IEC 60529 / EN 60529	Protection against accidental contact, foreign Objects and water

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**6. CT, PT Ratio and Transformer Details:**

CIRCUIT	33kV CT RATIO/CLASS
33kV Feeder	400-200/1-1 A; Class:0.5; 5P15; 10 VA, 10 VA
33kV side of 33/11kV transformer	400-200/1- 1-1A, Class:0.5/5P15/PS ; 10 VA, 10 VA, 10 VA
11kV side CT for Transformer	600-400/1-1-1A, Class:0.5/5P20/PS at phase side (Indoor Panel); 15VA; 30mA at V _k /2 (at lower ratio), V _k ≥170 V at LR
11kV transformer Bushing CT for REF	600/1A, PS for 10 MVA 33/11kV transformers for both Phases & neutral. 400/1A, PS for 6.3 MVA 33/11kV transformer for both phases and neutral.
33 kV PT	33 kV, single phase, Electromagnetic type
PT Ratio/Class	33kV/V ₃ , 110V/V ₃ -110V/V ₃ -110V/V ₃ , Class: 0.5/3P/3P; Burden:- 20VA/20 VA/ 5 VA
Transformer details	33/11kV, 6.3/10 MVA, Dyn11

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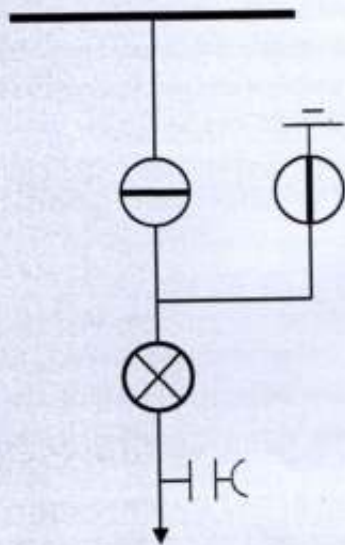
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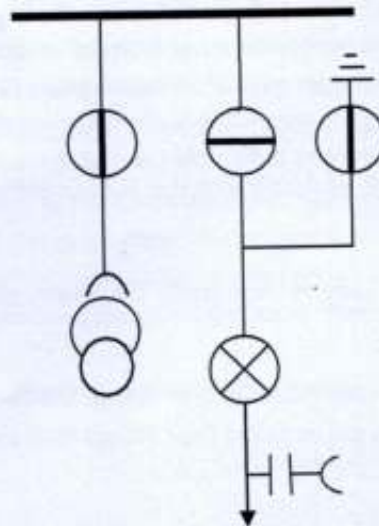
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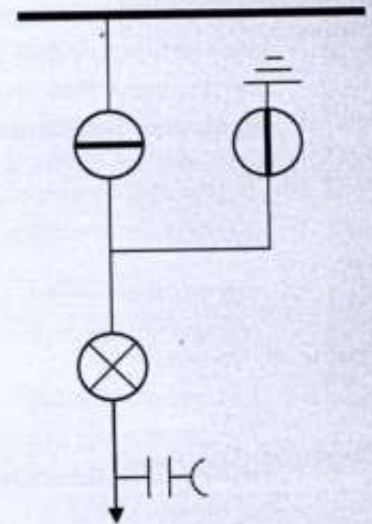
SLD for different combination of 33 kV GIS panel :



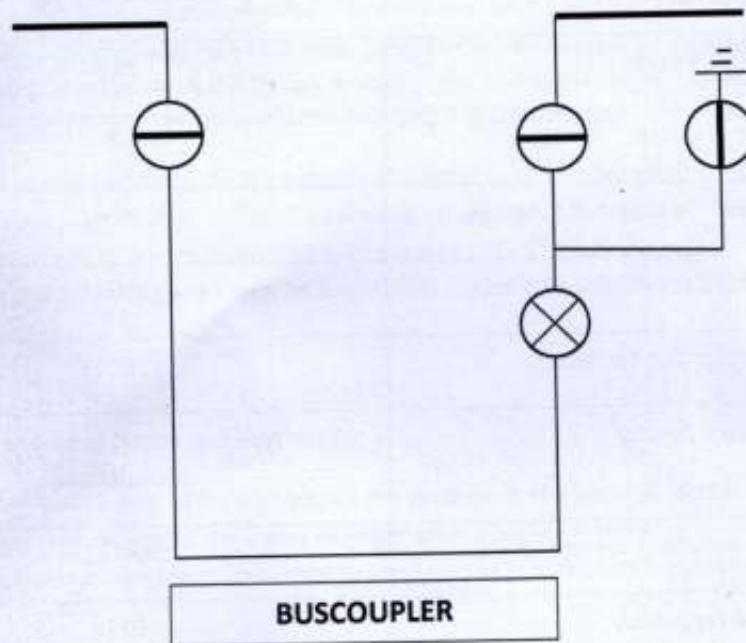
INCOMER/OUTGOING



INCOMER/OUTGOING + BUS PT



TRANSFORMER



BUSCOUPLER

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**8. Busbar and Insulators:**

- 8.1 33 kV busbar shall be housed in SF₆ gas chamber.
- 8.2 Busbar shall be of made of electrolytic high grade Copper of adequate size and bus bar size calculation/supporting type test report shall be submitted for approval (current density of copper shall not exceeded more than 1.6 Amp/sq.mm). They shall be adequately supported on insulators to withstand electrical and mechanical stresses due to specified short circuit currents.
- 8.3 Busbar shall be supported on the insulators such that the conductor expansion and contraction are allowed without straining the insulators.
- 8.4 Bus bar cross-section shall be uniform throughout the length of switchgear board. Bus bars shall be in SF₆ gas insulated compartment.
- 8.5 All compartment / piping for SF₆ gas including their fittings shall be made of non- magnetic stainless steel housings All compartment / piping for SF₆ gas including their fittings shall be made of non- magnetic stainless steel housings or high grade Aluminum alloy.
- 8.6 Busbar insulators shall be of arc and track resistant, high strength, non- hygroscopic, non- combustible type and shall be suitable to withstand stresses due to over-voltages, and short circuit current. In case of organic insulator partial discharge shall be limited to 20 pico coulomb at rated Voltage X 1.1/V³. The temperature of the busbars and all other equipment, when carrying the rated current continuously shall be limited as per the stipulations of relevant Standards, duly considering the specified ambient temperature (50°C). Clearances between phases and between phase and earth shall be kept liberally so as to obtain high reliability.

Bus bar System	SINGLE BUS BAR
Rated Voltage	36 kV
No of phases	3
Rated Power Frequency Withstand Voltage	70 kV
Rated Lightning impulse withstand Voltage	170 kV
Rated current for bus bar system	1600A
Current density	AS per type test report to be provided by the manufacturer
Rated Short time withstand current (3 Ph)	25kA/3 Sec
Rated Frequency	50 Hz

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Circuit Breaker :

Circuit breakers should be Vacuum type for 33 kV. They should be according to IEC 62271-100 standard. Rated operating sequence must be O-0.3sec-CO-3min-CO. Operating time for opening, breaking and closing, less than 100 ms. Circuit breaker must be E2/M2/C2 type (electrical, mechanical endurance and capacitor breaking/making ability) according to relevant standard. Circuit breaker operating mechanism will be outside the SF₆ tank, and electrically operated. The operating mechanism should consist in the following items. Spring system that stores the necessary energy for opening and closing operation. Spring charging system (motor operated) that automatically recharges the springs. Mechanical "charged-discharged" position indicator for CB opening and closing shall be provided.

The front of the operating mechanism shall be provided with a mimic fascia where the indicating elements shall be integrated directly to the mechanism

There shall be Manually operated spring charging system (in case of lack of auxiliary power supply).

Electrical system including:

- Closing coil
- Tripping coil (2 nos)
- Anti pumping relay
- Trip circuit supervision relay.
- Mechanical emergency trip push button.

Circuit breaker shall be provided with two trip coils. Suitable indicators shall be provided on the front of panel to indicate OPEN / CLOSED conditions of the circuit breaker, and CHARGED / DISCHARGED conditions of the closing spring, SF₆ gas density monitor for all gas compartments and trip Circuit supervisions indicating lamp.

The closing coil shall operate satisfactorily at all values of control supply voltage between 85-110% of the rated voltage. The trip coil shall operate satisfactorily under all operating conditions of the circuit breaker up to its rated short circuit breaking current at all values of control supply voltage between 70- 110% of the rated voltage. The trip coil shall be so designed that it does not get energized when its healthiness is monitored by indicating lamps and trip coil supervision relay.

Anti-pumping:

Function: If constant CLOSE and OPEN commands are present at the vacuum circuit-breaker at the same time, the vacuum circuit-breaker will return to the open position after closing. It remains in this position until a new CLOSE command is given. In this manner, continuous closing and opening (= pumping) is avoided.

10. Electrical parameters of the breakers:

1	Nominal System Voltage	33 KV
2	Highest System Voltage	36 kV(rms)
3	No. of Poles	3
4	No. of Trip Coil	2
5	Frequency	50Hz
6	Basic Insulation Level	170 kV(p), 70 KV (rms)
7	Operating Duty	O-0.3sec-CO-3min-CO
8	Total Break time for any current up to	3 cycles max (60ms)
9	Control Circuit Voltage	110 V DC +10% to-15%
10	Continuous Current rating at Ambient temperature (50°C) of	
	a) Incomer and Outgoing	1250 A
	b) Bus coupler	1250 A
11	Continuous Current rating at Ambient temperature (50° C) of Bus coupler	1250 A
12	Short Circuit Breaking Current	25 kA for 3 Sec

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11. Disconnecter and Earthing switch :

The three position (Isolator- OFF- Earth) Earthing and Disconnector switch will operate always at de-energized state and the making capacity is provided by the circuit-breaker, which is designed for these purposes. The Disconnector must be according to the latest edition of the IEC 62271-102 and shall meet requirements as specified in paragraph ratings. The indication of the position of the Disconnector must be mechanical. The Operating mechanism must be outside the SF₆ atmosphere and accessible from the front. The Disconnector must be of single rotation-driving axis both for the Disconnector and the Earthing switch. It is mandatory that the operation from "closed to busbar" to "ready to earth" is made in two separate operations, closed - open and open - ready to earth. These are completely independent operations, with two separated operating access.

Isolators or isolators combined with earthing switches (switches-Open, Close Earthed) shall be motor operated. In case of emergency, Manual operation shall be possible. The earthing position of all 3 phases shall be visible via a mechanical position indicator (MIMIC) directly connected to the drive shaft on panel front Fascia.

The interlocks shall be provided as per standards. It shall be possible to control these switches from front of the panel & remotely from SCADA through IED. Key interlocks shall also be provided for local manual operations.

12. Mechanical Safety and Interlocks:

- i) The switchboard must be fitted with all the interlocks needed to prevent incorrect operations which may jeopardize safety of personnel in charge of operating the installation, as well as the efficiency and reliability of the apparatus. Internal mechanical interlocks of the panel with the circuit breaker closed, the interrogation slide is locked for the disconnector and the earthing switch. (Restriction to the insertion of Hand Crank for Disconnector-Earth Switch when CB is ON).
- ii) The interrogation slide always release some insertion opening only (disconnecter or earthing switch), or both of them are locked. (To ensure that either Disconnector-or Earth Switch operating at a time). The crank for the disconnector and earthing switches can only be removed in its appropriate end position.
- iii) When the crank on the disconnector or earthing switch is still in place, or when the interrogation slide is open, the following components are locked:
 - ON push button of circuit breaker
 - ON pulse is interrupted
- iv) In particular, the following mechanical interlocks at least must be provided to prevent;
 - Operation in busbar if the circuit breaker is closed.
 - Closing of the earthing switch when busbar isolator are closed.
- v) The following interlocking dependencies must be provided-
 - Interdependence of disconnector with earthing switch in the same panel.
 - Interdependence of the circuit breaker and the earthing switch in the same panel.
 - Dependence of the disconnector / Circuit breaker on the bus coupler circuit breaker and vice versa.
- vi) The basic interlocking between disconnector switch and circuit breaker may be achieved by aux relays/other mechanical control equipment. The interlocking logic shall be operational and shall be fool proof and independent in the event of relay/BCU failure for operator's safety.
- vii) Gas Pressure Interlock with alarm (Not Tripping) must be there.

13. Earthing and Earthing Devices:

- a. The grounding system for GIS shall be designed and provided as per IEEE-80-2000 and CIGRE- 44 to protect operating staff against any hazardous touch voltages and electro- magnetic interferences.
- b. The earth busbar made of electrolytic high grade copper with cross sectional area of minimum 300 sq mm shall be provided at the bottom in all the panels and interconnected with adjacent panels in the panel board through a connecting link to form a common earth busbar for the entire panel board ready to connect to the substation earthing grid. It shall be welded to the framework of each panel and each breaker earthing contact bar. The earth bus shall have sufficient cross section (minimum 300sq mm) to carry the momentary short-circuit and short time fault currents to earth without exceeding the allowable temperature rise.
- c. Suitable arrangement shall be provided at each end of the earth bus for bolting to station earthing grid. All joint splices to the earth bus shall be made through at least two bolts and taps by proper lug and bolt connection.
- d. All non-current carrying metal work of the switchboard shall be effectively bonded to the earth bus. Electrical continuity of the whole switchgear enclosure frame work and the truck shall be maintained even after painting.
- e. All metallic cases of relays, instruments and other panel mounted equipment shall be connected to earth by independent stranded copper wires of size not less than 2.5 sq. mm. Insulation colour code of earthing wires shall be green. Earthing wires shall be connected to terminals with suitable clamp connectors and soldering shall not be acceptable. Looping of earth connections which would result in loss of earth connection to other devices, when a device is removed is not acceptable. However, looping of earth connections between equipment to provide alternative paths of earth bus is acceptable.
- f. PT and CT secondary neutral point earthing shall be at one place only on the terminal block. Such earthing shall be made through links so that earthing of one secondary circuit may be removed without disturbing the earthing of other circuits.
- g. The panel shall have Voltage Presence Indicator (VPI) to warn the operator against earthing of live connections.
- h. All hinged doors shall be earthed through flexible earthing braid.
- i. Separate earthing for switchgear and composite VCB panel shall be provided.

14. Instrument Transformers:

- a. The current transformers (CTs) and voltage transformers (VTs) should be outside the gas insulation and easily replaceable without any kind of GAS handling .
- b. All current transformers shall be ring type (epoxy/cast resin/tape wound) whereas voltage transformers (PT) shall be cast resin insulated type. Must provide details of ratio, output, class and accuracy for all CTs & PTs in its supply on the panel itself.
- c. Instrument transformers shall be suitable for continuous operation at the ambient temperature prevailing inside the switchgear enclosure , when the switchboard is operating at its rated load and the outside ambient temperature is 50° C. The class of insulation shall be E or better.
- d. The secondary windings of CTs & PTs shall be earthed at one point through a removable link, with provision for attaching test links.
- e. The current transformers, Inductive type, shall be in accordance to IEC 60185, IEC60044-1, IS 2705, IS 16227, read with latest revisions and the general requirements. The current transformers shall be core cast resin type and shall comply with the ratings indicated in the single line diagram. They shall be outside gas

compartment. The secondary wiring of 2.5 sq.mm. shall be taken out to the LV compartment via secondary stud.

- f. The Potential transformers shall be in accordance to IEC 60185, IEC 60196, IEC 60044-3, IS 3156, IS 16227 read with the latest revisions and general requirements as detailed in foregoing paras of specification. Easy access to Potential Transformers should be possible.
- g. Epoxy encapsulated / Metal enclosed Potential Transformers Single Pole insulated PTs which can be simply and safely changed shall be used exclusively. They shall be Plug in type with isolator and mounted outside the gas compartment for better accessibility and maintenance.
- h. Current transformers shall be rated to withstand the thermal and magnetic stresses resulting from the maximum available through fault current, the momentary and time rating requirements same as that of the switchgear.
- i. Must provide calculations together with current transformer magnetization curves to prove the through fault stability of the protection up to the switchgear fault rating.
- j. Secondary terminals of current transformers and potential transformers fitted in the panel shall be easily accessible without dismantling from the panel.
- k. Secondary terminals of current transformers and potential transformers shall be wired to disconnecting type terminal blocks, located at an accessible place in air filled low voltage compartment. The CT TBs shall be provided with short-circuiting links for each CT shorting and connection to earth. Each CT connection to the TBs shall be provided with ring type lugs. All wiring shall be colour coded.
- l. Use of voltage presence indicator sensors having high linearity over the entire working range and incorporating potential / capacitive divider circuit for connection to voltage indicator lamps.
- m. Nameplates shall be fixed in a position so that details can easily be read when fitted in the cubicle.
- n. Bus bar voltage transformers shall have provisions for safely disconnecting the fuses (at secondary side) and to avoid back feed of the transformers.
- o. PT Selection scheme must be there.
- p. All instrument transformer shall be outside the gas compartment.

a. P.T. SECONDARY CIRCUIT:

There may be two nos. 33 kV bus PT, one in each bus section. P.T. supply shall be available from selected 33 kV Bus P.T. through suitable PT selection scheme by switch. Two sets of Fuse and link of suitable rating shall be provided for the Incoming P.T. supplies and two sets, one for each PT of 3 nos. coloured LED indicating lamps shall be provided for supervision of the Fuse. Lamps shall be connected between respective phases and neutral. The arrangement of distribution of P.T. Secondary Circuit shall be as follows:

- (i) Potential supply to the protective relay circuit for Feeder where necessary shall be fed from selected Bus P.T. supply bus.
- (ii) Potential supply to meters, Energy meters and indicating instrument of each panel shall be fed from selected Bus P.T. supply bus.
- (iii) Selected P.T. secondary supply to the protective relays of each panel shall be fed through 4 poles - MCB and link in neutral in each panel where necessary with two change over contacts for annunciation.



- (iv) Selected P.T. secondary supply for metering and indicating instruments of each panel shall be through 4 pole MCB in each phase and link in neutral in each panel of 33kV system voltage.
- (v) PT selector switch PT -1/2 Stay put Type, 2 Position type, 16 Ways, minimum 16A shall be provided in each panel.

b.

CT Technical Parameters:**I. 33 kV Incomer/ Outgoing feeder panel**

Sl. No.	ITEM	Core-I	Core-II
i.	Purpose	metering	Protection
ii.	Primary Current rating	400-200 A	400-200A
iii.	Secondary Current rating	1Amps	1Amps
iv.	Rated Burden	10 VA	10 VA
V	Accuracy Class	0.5	5P15
Vi	Instrument Security factor / Accuracy Limit Factor	<=5 at LR	15

II. 33 kV outgoing Transformer panel:

Sl. No.	ITEM	Core-I	Core-II	Core-III
i.	Purpose	Metering	Protection	Special
ii.	Primary Current rating	400-200 Amps	400-200 Amps	400-200 Amps
iii.	Secondary Current rating	1Amps	1Amps	1Amps
iv.	Rated Burden	10 VA	10 VA	10 VA
V	Accuracy Class	0.5	5P15	PS Vk 70 V at LR, le Max 30 mA at Vk/2 at LR
Vi	Instrument Security factor / Accuracy Limit Factor	<= 5 at LR	15	
Vii	Class of Insulation	E or better		

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c. **PT Technical parameters:**

The voltage transformers shall be inductive metal-enclosed transformers.

Sl. No.	ITEM	33 KV
i.	Voltage class	36 KV
ii.	Rated primary voltage (kV rms)	33000/v3
iii.	Secondary voltage (volts)	Wdg-I & Wdg-II: 110/v3 Wdg-III: 110/3
iv.	Rated Burden	Wdg-I : 20 VA (metering) Wdg-II: 20 VA (protection) Wdg-III: 5 VA (to be used as open delta connection with suitable damping resistance)
V	Accuracy Class	Wdg-I: 0.5 Wdg-II: 3P Wdg-III: 3P
Vi	Class of Insulation	E or better
Vii	One minute power frequency withstand voltage for Secondary wiring (kV rms)	1.2 continuous and 1.9 for 30 seconds
Viii	Method for earthing the system	Impedance earthed

PT's for Busbar Voltage metering:

Location	Mounting
In Busbar Compartment	Plug in type with isolator Top mounted over the Busbar & directly connected to the Busbar system. No separate compartment is required.

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**15. CABLE TERMINATIONS:**

- a. Power cables shall be bottom entry, unless specified otherwise shall be provided. A socket/bushing and plug assembly shall be provided for at least two nos. of the power cables at each terminal, i.e. provision shall be made to connect at least two nos. of specified cable size may connect in each bay. Connections for power cables shall be according to DIN 47673, outer cone/ inner cone type as per relevant IEC.
- a. Cable terminating facilities and terminals shall be for the specified cable type, gland and conductor size.
- b. Terminal blocks shall be arranged and positioned to afford easy access for carrying out external cable termination, testing, inspection and maintenance. There shall be clear space allowed between the terminals and the gland-plate for the spreading land termination of external conductors.
- c. CT and PT secondary wire terminal block shall be as per wire phase colour.
 1. Red phase : Red colour
 2. Yellow phase : Yellow colour
 3. Blue Phase : Blue colour
 4. Neutral : Black colour
- d. The panel wiring shall be on one Side of the terminal block only. No more than two wires shall be connected to a terminal.
- e. Terminal blocks shall be shrouded or provided with transparent covers. Pinch screw type terminals are not acceptable.
- f. Terminals for different voltages shall be separated by partitions. CT wiring and control wirings shall not be on the same terminal stripe.
- g. A terminal box or chamber with un-drilled gland plate or entry panel of sufficient dimensions to terminate the specified cables shall be provided.
- h. Positioning of cable terminations shall avoid obstruction of other cable terminations; removable cover set and provide for easy access for terminating cables.
- i. Cable supports shall be provided (where practicable) to avoid undue strain on the cable termination.
- j. Separate terminal arrangements shall be provided for each panel and for power and control cables.
- k. All SCADA terminal shall be terminated at respective LV compartment in separate TB, SCADA terminal shall be BLUE in colour.
- l. Provision for control cable entry with gland plate shall also be provided in front of the panel. Control and power cable trench layout has been provided with this technical specification to comply with the panel design.

16. Low voltage Compartment (Instrument Chamber) :

The panels shall be with low voltage compartment consisting control switches, fuses, MCBs, Push Button, indication and metering instruments, TTB, protection relays, heater and other secondary equipment's. The SLD/ mimic shall be provided on the front fascia and not on the LV chamber.

The front side shall have Mimic as per single line diagram with control switches and mechanical and electrical 'Position indicators' circuit breakers, disconnectors and earth switches. Control switches/Push buttons shall be provided adjacent to respective equipment position indicators in Mimic for ON-OFF operation of circuit breakers, disconnectors and earth switches.

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The Communication device shall have USB/RJ 45 communication port in the front for the setting software use and at the back there has to be of this redundant communication port: 2xRJ-45/FO Ethernet port.

The Communication device shall support following communication protocols: Modbus TCP, Modbus RTU, Human-Machine communication, display and Human-machine communication, PC.

The IEC 61850 communication has to support peer to peer communication (GOOSE). The IEC 61850 interface has to be able to configure through setting software. Also the same setting software has to be able to generate ICD files.

Live line Indicators: - Capacitive voltage indicators shall be provided on feeder side in outgoing and incoming feeders to indicate the voltage presence in each phase and to prevent the closing of earth switch in case the part is live. It shall have sufficient output contacts for substation Automation System and interlock purpose.

Panel interlocking wiring cables shall be supplied with panels.

End termination required for cables should be touch proof type.

17. Constructional features:

- a) The Panel shall be suitable for erection of flush concrete floor and secured to it by means of evenly spaced grout bolt projecting through the suitable base frame steel channels members.
- b) The manufacturer shall ensure that the equipment specified and such unspecified complementary equipment required for completeness of protection/control scheme be properly accommodated in the panels without congestion.
- c) Panels shall be completely metal enclosed and shall be dust, moisture and vermin proof for tropical use.
- d) Panels shall be free standing, floor mounting type and shall comprise structural frames enclosed completely with specially selected smooth finished, cold rolled sheet steel of thickness not less than 2 mm. There shall be sufficient reinforcement to provide level surfaces, resistance to vibration and rigidity during transportation and installation.
- e) Design, material selection and workmanship shall be such as to result in neat appearance, inside and outside with no welds, rivets or bolt head apparent front outside, with all exterior surfaces true and smooth.
- f) All holes and extension windows in the Panel shall be blanked and access doors shall be lined with compressible liners at the edges. The WBSEDCL will shut off the bottom crevices with cream cement, the Cable Entry holes with weak concrete and the cable trench with present R.C. Slabs or checker plate. All control and supply cables will be laid in a distribution trench running under the panel. The Cable will branch off into each cubicle through entry holes in the concrete floor opening in the bottom cubicles. Necessary Drawings for concrete floor and trench shall be supplied by the manufacturer to enable the WBSEDCL to construct the foundation floor for these panels. The drawings shall show details of the distributing trench, cable entry holes, glands and positions of grouting bolts. The WBSEDCL will prepare foundation with pocket for grouting bolts. The manufacturer shall supply channel base, suitable grouting bolts, lock nut and washers.
- g) Control Cable entries to the panel shall be from the bottom. Bottom plates of the panels shall be fitted with detachable gland plates to allow cable entries from the bottom. Gland plates shall be suitable for fixing the cable glands at an elevated height of at least 100 mm above the ground level. Terminal Connectors and Test terminal blocks for cables shall be fixed at an elevated height of at least 200 mm above the Bottom plate. Side blocks cut out to be arranged at both sides of the panel for inter panel bus wires.

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h) The switch board shall have the facility for extension on both sides.

i) Gas Pressure relief device/Explosion Vent/Pressure Relief duct shall be provided for each SF6 gas compartment, so that in case of a fault in a compartment, the gases produced are safely vented out, thereby minimizing the possibility of its spreading to other compartments and panels. Both the gas filled chambers (busbar & VCB) will be discrete and will be equipped with individual Gas Pressure relief device/Explosion Vent/Pressure Relief duct.

17.1 General:

a) Materials shall be new; the best quality of their respective kinds and such as are usual and suitable for work of like character. All materials shall comply with the latest issues of the specified standard unless otherwise specified or permitted by WBSEDCL.

b) Workmanship shall be of the highest class throughout to ensure reliable and vibrations free operations. The design, dimensions and materials of all parts shall be such that the stresses to which they may be subjected shall not cause distortion, undue wear, or damage under the most severe conditions encountered in service.

c) All parts shall conform to the dimensions shown and shall be built in accordance with approved drawings. All joints, datum surfaces and meeting components shall be machined and all castings shall be spot faced for nuts. All machined finishes shall be shown on the drawings. All screw, bolts, studs and nuts and threads for pipe shall conform to the latest standards of the International Organization for Standardization covering these components and shall all conform to the standards for metric sizes.

d) All materials and works that have cracks, flaws or other defects or inferior workmanship will be rejected by WBSEDCL.

17.2 Assembly:

Necessary items of equipment shall be assembled in the factory prior to shipment and routine tests shall be performed by the manufacturer as per the requirements of the latest issue of IEC/IS as specified under each equipment in these specifications to demonstrate to the satisfaction of WBSEDCL that the switchgear panels comply with the requirements of the relevant IEC/IS standards.

17.3 Casting:

Casting shall be true to pattern, of workmanlike finish and of uniform quality and condition, free from blowholes, porosity, hard spots, shrinkage defects, cracks or other injurious defects, shall be satisfactorily cleaned for their intended purpose.

17.4 Welding:

Wherever welding is specified or permitted, a welding process, including stress relieve treatment as required if necessary, conforming to an appropriate and widely recognized professional standard shall be used. All welders and welding operators shall be fully qualified by such a standard.

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17.5 MOUNTING:

- All equipment on and inside the panels shall be mounted and completely wired to the terminal blocks ready for external connection.
- Equipment shall be mounted such that removal and replacement can be accomplished individually without interruption of service to adjacent devices and are readily accessible without use of special tools. Terminal marking shall be clearly visible and of permanent nature.
- The manufacturer shall carry out cut-out, mounting and wiring of the bought out items which are to be mounted in the panel in accordance with the corresponding equipment manufacturer's drawings.
- The centre line of switches, push buttons and indicating lamps shall be maintained a certain height from the bottom of the panel for easy operation of the operator at standing position. The centre line of the numerical relays and meters and recorders shall be not less than 450 mm from the bottom of the panel.
- The centre lines of switches, push buttons and indicating lamps shall be matched to give a neat and uniform appearance. Likewise the top of all meters, relays and recorders etc. shall be in one line.
- The location of the switches shall be within working height from the floor level for easy and comfortable operation.
- All the equipment connections and cabling shall be designed and arranged to minimise the risk of fire and damage.

17.6 WIRING:

17.6.1 All wiring shall be carried out with 1100 volts grade single core, multistrand flexible tinned copper wires with PVC insulation which has provided its utility in tropical region against hot and moist climate and vermin (Misc. white ant and cockroaches etc.) Rubber insulated wiring will not be accepted. Wire numberings and colour code for wiring shall be as per IS: 5578/1984. The wiring should be encased in suitable width PVC casing. The PVC insulation shall be of FRLS category, flame vermin & rodent proof.

17.6.2 The sizes of wiring in different circuit shall not be less than these specified below:

TABLE-I

Circuit	Permissible size of wire
Metering and Relaying Circuits connected Current Transformer including earthing	2.5 mm ²
Potential Circuits for metering and Relaying, Control, Visual Audible Alarms and Signalling Circuit	1.5 mm ²
All interpanel Buswire and earth wire	2.5 mm ²



17.6.3 The following colour schemes shall be used for the Wiring:

TABLE - II

Circuit where used	Colour of Wire
Red Phase of Instrument Transformer Circuits	Red
Yellow Phase of Instrument Transformer Circuits	Yellow
Blue Phase of Instrument Transformer Circuits	Blue
Neutral connection, earthed or not earthed in the instrument Transformer Circuit	Black
A.C. Control Wiring Circuits using auxiliary supply and	Black
D.C. Control Wiring Circuit using Battery Supply	Grey
Earth Connection	Green

- All internal wiring shall be securely supported, neatly arranged, readily accessible and connected to equipment terminals and terminal blocks. Wiring gutters & trough shall be used for this purpose.
- Longitudinal troughs extending throughout the full length of the panel shall be used for inter panel wiring. Inter connections to adjacent panels shall be brought out to a separate set of terminal blocks wires.
- Wiring connected to the space heaters in the cubicles shall have porcelain beaded insulation over a safe length from the heater terminals.
- Wire termination shall be made with solder less crimping type and tinned copper lugs which firmly grip the conductor and insulation. Insulated sleeves shall be provided to all the wire terminations. Engraved core identification plastic ferrules marked to correspond with panel wiring diagram shall be fitted at both ends of each wire. Ferrules shall fit tightly on the wire and shall not fall off when the wire is disconnected for any purpose. Termination shall be such that no strand of a conductor shall left loose or overhanging. Conductor termination shall be secured to the holding nuts/screws, terminal blocks etc. with washers interposed between the terminals/holding nuts/screw heads. The terminals shall be so connected that no conductor ferrule code gets masked due to overlay of conductors.
- All spare contacts of relays shall be wired up to terminal blocks.
- Each wire shall be continuous from end to end and shall not have any joint within itself individually.
- Wires shall be connected only at the connection terminals or studs of the terminal blocks, meters, relays, instruments and other panel devices. Terminal Ends of all wires shall be provided with numbered Ferrules. At point of inter-connection where a change of number is necessary, duplicate Ferrules shall be provided with the appropriate numbers on the changing end.
- At the terminal connection, washers shall be interposed between terminals, wire terminals and the holding nuts. All holding nuts shall be secured by locking nuts. The connection stud shall project at least 6 mm from the lock nut surface. Wire ends shall be so connected at the terminal studs that no wire terminal numbered

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ferrule gets masked due to succeeding connections. All wires shall be suitable for bending to meet the terminal stud at right angles with the stud axis, and they shall not be skewed.

- i) All studs, nuts, bolt's screws etc. shall be threaded according to the British Standard practice unless WBSEDCL's prior approval to any other practice of threading is obtained.

18. TERMINAL BLOCK CONNECTION:

Terminal blocks shall be of clip-on design made out of non-trackable insulating material of 1100 V grade. All terminals shall be stud type, with all current carrying and live parts made of tinned plated brass. The studs shall be of min 4 mm dia brass. The washers, nuts, etc. used for terminal connectors shall also be of tinned plated brass. All blocks shall be shrouded by easily removable shrouds made of transparent die-electric materials.

The terminal connector/blocks shall be disconnecting type terminal connectors for PT and same with automatic shorting of C.T. secondary terminals shall be provided in CT secondary circuit. All other terminal connectors shall be Non-disconnecting type. Terminal should be shock protected in single moulded piece. Terminal block should have screw locking design to prevent loosening of conductor. Provision shall be made on each pillar, for holding 10% extra connection (5% incoming + 5% outgoing).

At least 20% spare terminals for each type shall be provided. All terminals shall be provided with ferrules indelibly marked or numbered and identification shall correspond to the designations on the relevant wiring diagrams. The terminals shall be rated for adequate capacity which shall not be less than 10 Amps for control circuit. For power circuit it shall not be less than 15 Amps.

19. SPACE FOR CONTROL CABLES AND CABLE GLANDS:

Sufficient space for receiving the Control Cables inside the Panel at the bottom of the cubicles and mounting arrangement for the terminal cable glands shall be provided. Removable type separate cable entry plate (may be two) shall be fixed with bottom plate. The specification does not cover supply of control cables and cable glands for which the WBSEDCL will make separate arrangement.

20. SPACE HEATERS:

240 V, 50 Hz Tubular Space Heaters suitable for connection to the Single Phase A.C. Supply complete with On-Off Switches located at convenient position shall be provided in each compartment of the Panel to prevent condensation of moisture. The Watt loss per Unit surface of heater shall be low enough to keep surface temperature well below sensible heat. A thermostat control unit with variable temperature shall be installed to control the heater. The 240 V AC supply for the heater shall be controlled by a suitably rated single pole miniature circuit breaker compartment to be mounted on an insulator.

21. DISTRIBUTION AND CONTROL OF AUX. POWER CIRCUIT

a. D.C. CIRCUIT:

There shall be only one 110 V D.C. for the entire Panel fed from a D.C. Distribution Panel. A continuous D.C. Bus shall be provided in the Panel and D.C. supply for control, protection, indication and supervision of circuit breaker and other equipment shall be teed off from D.C. bus through a set of 20 Amp rated H.R.C. Fuse on positive and negative side. D.C. supply to be teed off shall be distributed within the Panel as below:

Control DC scheme both positive and negative side with 16 Amp fuse

Trip Ckt 1 and Trip Ckt 2 without fuse (link to be used); closing circuit with 10A fuse.



Indication Circuit through a set of 6 Amp. HRC Fuse both at +ve and -ve side

Protective relay circuits through 6A fuse both at +ve and -ve side

Annunciation ckt with 6 Amp fuse on both at +ve and -ve side

DC Emergency Lamp with 6 Amp fuse both at +ve and -ve side

One nos. D.C. operated no-volt auxiliary relay (self reset type) shall be provided in each panel with hand reset type flag with inscription — 'Main D.C. Fail', with 4NO+4NC (including one NC type coil cut off contact) contacts. 2 NC contact for 'DC fail' alarm and Indication, 1NO wired upto SCADA TB and 1NO wired up to spare TB.

A.C. operated no volt auxiliary relay (self reset type) rated for 240V shall be provided with hand reset flag with inscription — 'DC Fail Accept' with 4NO+4NC (including one NC type coil cut off contact) contacts.

One Push button having N/C Contact used in Series with the above relay for 'D.C. Fail Test' purpose.

b. A.C. CIRCUITS:

240 Volts, Single Phase A.C. Aux. Supply to the Control and Relay Panel will be fed from A.C. Distribution Panel through a 16Amp MCB provided there. One 16 Amps rated HRC Fuse shall be provided at the Panel for the Incoming A.C. Supply. A.C. operated no volt auxiliary relay (self reset type) rated for 240V shall be provided in BC panel with hand reset flag with inscription — 'A.C. Fail' with 4NO+4NC (including one NC type coil cut off contact) contacts. One push button having N/C Contact used in Series with above relay for — 'A.C. Fail Test' purpose.

C. FUSE AND LINK:

Fuses shall be of cartridge type. Carrier and base for the fuse and links for all D.C. and A.C. Circuits shall have imprint of rating, voltage and circuit designation.

22. MIMIC DIAGRAMS:

i) Provision shall be made for 10 mm. wide mimic diagram on the exterior of the front panel to represent the single line arrangement of the station equipment. Provision shall be made in such a way that centre line of the mimic bus shall be at a suitable height from the bottom of the Panel.

ii) Colour scheme for mimic diagram as follows:-

kV Class	Colour	Shade Index as per ISS
33 kV	Brilliant green	221
11 kV	Air Craft blue	108

iii) In 33 kV GIS panels, Symbol marking for the position indication of isolators, earth switches etc, ON/OFF indication for Circuit breaker, PT supply indication, CB spring charge, auto trip, trip ckt healthy etc. shall be mounted along the mimic diagram at appropriate location. Non-Discrepancy type control switch for the C.B. shall be mounted within the mimic, indicating the C.B. ON/OFF status.

23. LABELING:

All front mounted as well as internally mounted items including MCBs shall be provided with individual identification labels. Labels shall be mounted directly below the respective equipment and shall clearly indicate the equipment designation. Labelling shall be on aluminium anodised plates of 1 mm thickness, letters are to be properly engraved.

24. EARTH BUS :

Each panel shall be provided with continuous earth bus of specified size. The earth bus shall be of tinned plated copper, and all metallic cases of relays, instruments etc. shall be connected to this earth bus independently for their effective earthing. The wire used for earth connections shall have green insulation.

25. CONTROL SWITCHES: CIRCUIT BREAKER/Isolator/Earth Switch:

- (i) PISTOL GRIP TYPE Non- discrepancy T-N-C spring return type, non-lockable switch shall be provided for operation of circuit breaker. Similar type of switch O-N-C, but lockable shall be used for Isolator and earth switch. The switch shall be mounted in the mimic diagram itself such that the stay-put ('N') position will render the continuity of the mimic. One green LED for 'breaker open' indication and one red LED for 'breaker closed' indication shall also be provided adjacent to the T-N-C switch. Anti pumping relay will work together with the TNC switch.
- (ii) Switches should have finger touch proof terminals. For the convenience of maintenance, screw driver guide should be from top/bottom of the switch and not from the side. Terminal wire should be inserted from the side of the switch terminal.
- (iii) Terminal screws must be captive to avoid misplace during maintenance.
- (iv) Switch shall be escutcheon plate marked with Trip & Close.
- (v) Trip-neutral-close, with pistol grip handle must be pushed in to spring return to either trip or close position from Neutral position for safety and not just turn to trip.
- (vi) One contact to close in each position of Trip and Close. Contact rating shall be 12 A at 110 V DC.
- (vii) One NAC contact shall be used for Auto trip indication.
- (viii) One spare contact is required in off & on position.

26. LOCAL/SCADA SWITCH:

Local/ SCADA Switch, 2 Position Type, lockable, Stayput type, 10 ways, 25 Amp. Shall be used for and isolator operation except earth switch.

27. INDICATING LAMPS:

L.E.D. Type Indicating Lamps shall be provided on the Panel to indicate the following:

Sl. No.	Functions	Quantity	Colour of Lamp
1	C.B. Spring charged indication	1 No.	Blue
2	C.B. trip Coil/Circuit healthy indication	2 Nos.	White
3	C.B. Auto tripped indication	1 No.	Amber
4	Panel D.C. Fail indication	1 No.	Amber
5	Panel A.C. Fail indication (BC panel only)	1 No.	Amber
6	P.T. Supply indicating Lamp	2 sets	Red/Yellow/Blue
7	C.B. —ON indication	1 No.	Red
8	C.B. —OFF indication	1 No.	Green
9	Isolator ON	1 no	Yellow
10	Isolator OFF	1 No.	Green
11	Earth switch ON	1 no	Yellow
12	Earth switch OFF	1 No.	Green

28. CONTACT MULTIPLIER:

Contact multiplier relay may be incorporated if required during finalization of the scheme.

29. TERMINAL BLOCK / TTB :

3-Phase, 4-Wire Link type Test Terminal Block having sealing provision shall be provided in Metering Circuit of each Panel. Screw type TTB will not be accepted.

30. SAFETY EARTHING :

Earthing of metallic parts or metallic bodies of the equipment on the Panel shall be done with soft drawn single conductor bare Copper Tail connections shall have minimum area of 16 sq. mm. and the main earthing connection 60 sq. mm. These wires shall be connected by suitable terminals and clamps junction. Soldered connections shall not be employed.

The neutral point of star connected LV winding of instrument transformers and one corner of the open delta connected LV side of instrument transformers shall be similarly earthed by tail connected with main earth wire of Panel Earthing System. Multiple earthing of any instrument transformer circuit shall be avoided.

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31. PANEL LIGHTING:

1. The Panel LV compartment's interior shall be illuminated by LED lamps connected to 240 Volt Single Phase A.C. The illumination of the interior shall be free from shadows and shall be planned to avoid any strain or fatigue to the wire man likely to be caused due to sub-normal or non-uniform illumination.
2. A toggle switch or door operated switch shall be provided for control of A.C. lighting in each panel LV compartment.
3. One combined 15 Amps 3-Pin and 5 Amps 2-Pin Power Socket outlet together with Plus Pins shall be provided at convenient points in each BC Panel for A.C. Supply. All A.C. terminals of switches & sockets shall be covered to avoid electric shock.

32. ANNUNCIATOR:**ELECTRONIC ANNUNCIATOR**

- 32.1 Suitable Multi-way Microprocessor based electronic Annunciator for the visual and audible alarm on the panel using bright LEDs shall be provided in each panel to indicate over current and earth fault protection operated. In addition to above, each electronic annunciator of Transformer Panel shall have provision to indicate Transformer trouble trip/alarm function operated. Also one window of the Annunciator shall have to be used for Non-Trip A.C. Fail Alarm Indication and one window for Trip Circuit unhealthy indication. Each Electronic Annunciator shall have provision for connection with accept/reset/lamp test/mute Push buttons for proper functions. Electronic annunciator shall have provision for connection with Electronic Buzzer/Electronic Bell for Trip & Non-Trip Audio Alarm of common annunciation scheme. Electronic Annunciation shall have provision for flashing illuminating display with inscription for operation of respective Protection Relay. The Micro-Processor based Electronic Annunciator should have separate colored windows for Trip & Non-Trip Annunciation for easy detection.
- 32.2 Annunciator fascia units shall have translucent plastic windows for each alarm point.
- 32.3 Dedicated window for "DC fail" and "AC fail" with non trip alarm shall be provided in the annunciator.
- 32.4 Electronic Annunciator shall have first Fault Indication Facilities & System Watch Dog
- 32.5 Alarm inscriptions shall be engraved on each window in not more than three lines The inscriptions shall be visible only when the respective facia LED will glow.
- 32.6 Annunciator facia units shall be suitable for flush mounting on panels. Replacement of individual facia inscription plate and LED shall be possible from front of the panel.
- 32.7 Unless otherwise specified, one alarm buzzer meant for non-trip alarms and one bell meant for trip alarms shall be provided in each BC panel (mounted inside).
- 32.8 Each annunciator shall be provided with inbuilt 'Accept', 'Reset', 'Test' and mute push buttons, in addition to external panel mounted 'Accept', 'Reset' push buttons.
- 32.9 Special precaution shall be taken by the manufacturer to ensure that spurious alarm conditions do not appear due to influence of external magnetic fields on the annunciator wiring and switching disturbances from the neighboring circuits within the panels.
- 32.10 In case 'RESET' push button is pressed before abnormality is cleared, the LEDs shall continue to glow steadily and shall glow out only when normal condition is restored.

- 32.11 Any new annunciation appearing after the operation of 'Accept' for previous annunciation, provide a fresh audible alarm with accompanied visual alarm, even if the process of "acknowledging" "resetting" of previous alarm is going on or is yet to be carried out.
- 32.12 Provision for testing healthiness of visual and audible alarm circuits of annunciator shall be available.

16 Window Annunciation Scheme for Transformer Panel to indicate following functions:-

i)	Differential protection(87) operated	1 no.
ii)	Non-directional protection (O/C+E/F) operated	1 no.
iii)	Oil Temp./Winding Temp/MOG Alarm for transformer	1 no.
iv)	Oil Temp./Winding Temp Trip for transformer	1 no.
v)	REF 64R(HV side) tripped	1 no.
vi)	REF 164R(LV side) tripped	1 no.
vii)	Buchholz Alarm for transformer	1 no.
viii)	Buchholz Trip for transformer	1 no.
ix)	OLTC Buchholz/ Main Tank PRV Trip for transformer	1 no.
x)	AC fail	1 no.
xi)	Trip Circuit 1 Unhealthy	1 no.
xii)	Trip Circuit 2 Unhealthy	1 no.
xiii)	Non-directional O/C & E/F Relay Trouble	1 no.
xiv)	Differential relay trouble	1 no.
xv)	Panel DC fail	1 no.
xvi)	O/C and E/F relay trouble	1 no.

Mounting	Flush
No. of facia windows	16
Supply voltage	110 V DC and 240 V AC
No. of LEDs per window	2 (Min)
Lettering on facia plate	Properly engraved

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12 Window Annunciation Scheme for IC Feeders to indicate following functions:-

i)	Directional O/C operated	1 no.
ii)	Directional E/F operated	1 no.
iii)	Panel DC Fail	1 no.
iv)	Trip Circuit 1 Unhealthy	1 no.
v)	Panel AC fail	1 no.
vi)	Trip Circuit 2 Unhealthy	1 no.
vii)	Directional O/C & E/F Relay Trouble	1 no.
viii)	PT MCB Tripped	1 no.
ix)	Spare	1 no.
x)	Spare	1 no.
xi)	Spare	1 no.
xii)	Spare	1 no.

Mounting	Flush
No. of fascia windows	12
Supply voltage	110 V DC and 240 V AC
No. of LEDs per window	2 (Min)
Lettering on fascia plate	Properly engraved

12 Window Annunciation Scheme for OG Feeders to indicate following functions:-

i)	O/C operated	1 no.
ii)	E/F operated	1 no.
iii)	Panel DC Fail	1 no.
iv)	Trip Circuit 1 Unhealthy	1 no.
v)	Panel AC fail	1 no.
vi)	Trip Circuit 2 Unhealthy	1 no.

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vii)	O/C & E/F Relay Trouble	1 no.
viii)	PT MCB Tripped	1 no.
ix)	Spare	1 no.
x)	Spare	1 no.
xi)	Spare	1 no.
xii)	Spare	1 no.

Mounting	Flush
No. of facia windows	12
Supply voltage	110 V DC and 240 V AC
No. of LEDs per window	2 (Min)
Lettering on facia plate	Properly engraved

12 Window Annunciation Scheme for BC Panel to indicate following functions:-		
i)	Panel DC Fail	1 no.
ii)	Trip Circuit 1 Unhealthy	1 no.
iii)	Panel AC fail	1 no.
iv)	Trip Circuit 2 Unhealthy	1 no.
v)	Spare	1 no.
vi)	Spare	1 no.
vii)	Spare	1 no.
viii)	Spare	1 no.
ix)	Spare	1 no.
x)	Spare	1 no.
xi)	Spare	1 no.
xii)	Spare	1 no.

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Mounting	Flush
No. of facia windows	12
Supply voltage	110 V DC and 240 V AC
No. of LEDs per window	2 (Min)
Lettering on facia plate	Properly engraved

33. METERING PROTECTION, CONTROL AND INDICATION :

Meters and relays shall be capable of withstanding, without damage the secondary currents associated with the maximum available through fault current same as switchgear rated fault current.

Protection and metering devices shall be flush mounted, where possible shall be withdrawable. Protective main relays shall be of the numerical based programmable type with serial communication facilities.

Protective relays shall have electrical-reset facilities and clear operation indication, e.g. light emanating diodes. Non-tripping relays may be self-resetting if an indication of operation remains until hand reset.

Relay shall have front RJ 45/USB port for communication to a local PC with rear RJ45/FO port for connection to remote operation on IEC61850 protocol without any external convertor.

Relay shall have circuit breaker failure protection built-in.

Trip Relay shall have electrical reset facility.

Relay shall have under and over voltage protection for Incomer Panels.

Indication lamps (green-open, red-closed, amber-auto tripped, white-trip circuit healthy ircuit Earthed, blue-spring charged) shall be provided for each circuit breaker. Status of spring charging mechanism (Charged or Discharged) and positive drive mechanical position indicators, visible without opening the cubicle door, shall be provided for each circuit breaker. Volt-free contacts for remote indication shall be provided. Indications may be incorporated in the multifunction relay via relay LEDs.

Control of circuit breakers shall be carried out from a control switch at the low voltage compartment door. Provisions shall be implemented at the switchgear for breaker control from remote. Manual trips via relays are not acceptable.

34. Metering Scheme:

34.1 INDICATING INSTRUMENT AND METERS

- All instruments shall be flush mounted, back connected type and provided with dust tight cases for tropical use with dull black enamel finish. All fixing screws, nuts and threaded parts shall be designed to Indian Standards.
- All instruments shall be of class 0.5 type. The calibration of the instruments shall function satisfactorily when mounted on steel panels or alternatively magnetically shielded instruments shall be used.



- c. Instruments shall be capable of indicating freely when operated continuously at any temperature 0 to 50 degree C.
- d. All circuits of instruments shall be capable of withstanding applied load of 20% greater than the rated capacity for a period of eight hours.
- e. The instruments shall be capable of withstanding the effect of shock vibration and a di-electric test of 2000 Volts r.m.s. to ground for one minute as per relevant ISS.

34.2 Ammeters:

All ammeters shall be provided with selectable CT ratio. Individual ammeter for each phase shall be used to measure direct current without any selector switch. The ammeters to be supplied shall be "DIGITAL" type. The auxiliary power of the ammeters should be 240V AC. Multifunction meter may be used instead of ammeter.

34.3 Voltmeters :

Digital type Volt Meter shall be provided with direct reading scale. The maximum value of the volt-scale be 15% in excess of the normal Circuit Voltage. The rated secondary voltage of the Volt Meter shall be 110V A.C. However, the voltmeters to be supplied shall be "DIGITAL" type. The auxiliary power of the voltmeters should be 240V AC. Digital type voltmeter shall have 3 line display so that all ph to ph voltage can show at a time without any voltmeter selector switch. Multifunction meter may be used instead of voltmeter.

34.4 PT Selector Switch:

One PT selector switch, 2 position, stay put type shall be provided. PT selector switch to be used for selection of BPT 1 or BPT 2. One switch with sufficient nos. of pole may be used for both metering and protection winding.

34.5 Energy Meters:

Tariff Metering Equipment

(a) Three element Tri-vector Meters **shall be supplied by the WBSEDCL**. But Panel Wiring for the Meters along with Test Terminal Block and space for the Tri-vector Meters are to be provided for all the Panels except BC panel

35. NAME OF IDENTITY PLATES:

- a) All instruments, relays and such other similar electrical devices mounted on the panel shall be provided with name plates bearing the manufacturer's name, serial identifying number and the Electrical rating data, " Property of WBSEDCL", Guarantee :5 Years, scheme name, PO/LOA no. year of manufacture etc. .

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- b) 3 mm thick and 25mmX150mm brass or plastic plates bearing suitable identification marks shall be fixed under the terminal wiring at the test blocks, at the fuse blocks and at the cable terminals. Similar plates shall be fixed on the exterior of the panel in appropriate places to indicate function of control switches, push button etc. such as isolator control switch, breaker control switch, DC fail test, accept reset etc. Suitable identification marks shall be provided for individual casing part of the relays and other equipment. Plates should be screwed and riveted to the Panel.

36. PAINTING:

Panel painting shall be done by the modern process of painting. All unfurnished surface of the steel panel and frame work shall be sand blasted or suitably cured to remove rust, scale, foreign adhering matter or grease. A suitable rust resisting primer shall be applied on the interior and exterior surface of steel, which shall be followed by application of an undercoat suitable to serve as base and binder for finishing coat.

Details of Painting:-

Surface treatment	by seven tank process
Paint type	Powder coated. Pure polyester base grade A structure finish
Paint shade	RAL 7032 for external & internal surface
Paint thickness	Minimum 60 microns

37. Protection Scheme:

a. RELAYS:

i. GENERAL REQUIREMENT

The main protective relays SCADA Compatible Numerical Differential Relay, Numerical Directional O/C & E/F Relays shall be IEC 61850 compatible.

Credential for offered relays shall also to be furnished along with GIS panel in tender documents as pre-requisites of tender.

All numerical relays shall be provided with 'Relay Failure Annunciation contact'.

Live demonstration of any offered relay has to be arranged by the party at the laboratory of Distribution Testing Department of WBSEDCL, if felt necessary by WBSEDCL before finalizing the Bid.

ii. SCADA COMPATIBLE NUMERICAL DIRECTIONAL O/C & E/F RELAYS & DIFFERENTIAL RELAYS

The primary requirements of the relays are to protect the respective single circuit or double circuit feeders and 33/11 kV Power Transformers in the event of fault. Numerical relays shall provide suitable sensitivity for limited earth fault current.

The relay should be suitable for substation automation, primary circuit breaker operation through SCADA from remote control room.

All numerical relays shall be cyber security tested from CPRI, India; in line with the guidelines of the Ministry of power, Govt. of India.

THE DETAILED SPECIFICATION OF Directional O/C and E/F RELAY IS AS PER ANNEXURE-A OF SPECIFICATION

THE DETAILED SPECIFICATION OF DIFFERENTIAL RELAY IS AS PER ANNEXURE-B OF SPECIFICATION

iii. OTHER PROTECTIVE RELAYS

- Master Trip & other Auxiliary relays (other than the Numerical relays with IEC 61850 communication protocol) not previously supplied to WBSEDCL but having the Technical features as per technical specification may also be acceptable provided the performance certificate of those relays in any other power utility for a period not less than 01(One) year.

iv. OTHER PARTICULARS RELATED TO ALL RELAYS

- a. All shall conform to the requirement of IS: 3231 / IEC 255 and shall be suitable for operation within a temperature range 0°C to 55°C and 95% relative humidity. Relays shall be suitable for flush / semi flush mounting on the panel with connections from the rear, protected with dust tight cases for tropical use and with transparent cover removable from the front.
- b. All A.C. relays shall be suitable for operation at 50Hz. The current coils shall be rated for a continuous current of 1 amp and the voltage coil for 240V normal. The contacts of the relays shall be properly designed to prevent or minimise damage due to arcs which have to be broken successfully against 110V +/- 10% volt DC. When open, the contacts shall withstand a voltage of 115% of the normal circuit voltage. The relays shall be designed for satisfactory operation between 70% to 110% of rated D.C. voltage of the sub-station. The voltage operated relays shall have adequate thermal capacity for continuous operation.
- c. The timer shall have built in feature in the Numerical relays as mentioned in the specifications.
- d. The Relays shall preferably be provided with suitable Seal-in-Devices. Relays should be immune to all types of external influences like Electro static, Electromagnetic, Radio interference, shock etc.

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- e. All the numerical relay should have provision for setting all the features available in the relay and viewing those setting as well as different other parameters through both built in display unit as well as through PC. All numerical relays shall have self monitoring feature with watch dog contact. The supply of relay should be inclusive of necessary licensed software and hardware for interfacing with a PC, to be supplied by the bidder.
- f. External push button shall be used for reset of each numerical relay through relay's binary input and electrical reset of Master Trip relay.

V. PROTECTION SCHEMES

E-1-1 PROTECTION SCHEMES FOR 33 kV LINE PROTECTION

DIRECTIONAL PROTECTION

Directional O/C & Directional Instantaneous E/F Relays shall be required for 33 kV feeders as specified in the schedule of requirement. Each Feeder shall be provided with 3 elements IDMT Voltage polarized O/C Relays and single element voltage polarized E/F Relay. The O/C Relays shall be IDMT type with high set element. The E/F Relay shall have directional sensitive E/F setting having wide range of setting (1-40%) & wide range of definite time setting range minimum. 0.1 to 10 Sec. The relay shall also have instantaneous unit. The relay shall have necessary P.T. fuse failure monitoring scheme. **The Directional O/C & E/F Relay should have the Site-selectable provision for operating in both Directional & Non-directional mode depending on site requirement.**

Characteristics:-

O/C Element: IDMT with High Set Unit	Current Settings & Operating time	IDMT-50-200%, 0-3 sec. Inst.- 500-2000% or 400-1600%
MTA	Selectable MTA for Directional Relay should cover 1 st quadrant in a non-effectively grounded system	
Polarized P.T. Voltage	110 V A.C.	
E/F Element		
Current Setting	1-40% (minimum.) in very small steps	
Operating Time of Relay	Instantaneous	
Operating Time of Timer	0.1 to 10 Sec in very small steps	

MTA	Selectable MTA for Directional Relay should cover 1 st quadrant in a non-effectively grounded system	
Open Delta P.T. Voltage	63.5 V A.C.	

The numerical directional relay shall have in-built feature for derivation of zero sequence voltage internally.

E-1-2: One Directional protection relay (as specified in ANNEXURE-A) shall be used in Bus-coupler panel only to obtain the facility of relay Synchro check function for paralleling of two incomer feeder in Auto mode. Closing circuit interlock to be provided with one output contact in auto mode. One Auto/Manual switch shall also to be used in closing circuit for that purpose. In manual mode synchro check contact will be bypassed.

E-2 PROTECTION OF 33 kV INDIVIDUAL TRANSFORMERS

For protection of H.V. Side of the Transformers, following main protective relays are required

- Numerical O/C and E/F protection as specified in ANNEXURE-A .
- HV and LV side Restricted E/F protection to be provided by the two separate REF protection element in the numerical differential relay. Stabilising resistance and metrosil shall be used for both the REF protection scheme.
- Numerical Differential Relay shall be provided for transformer protection ANNEXURE-B.

Differential Relay shall be-

- Provided at 33kV panel of the transformers to be protected. It shall be numerical adjustable/variable percentage biased type differential relay.

Necessary software, cables, connectors and other accessories as required for download, analyze data etc. shall be within the scope of successful bidder.

- The relay shall be very fast in operation with an operating time less than 40 millisecond at 5 times setting.
- The relays shall be inherently stable for external through fault conditions without affecting the speed of operation for internal faults.
- The relay shall have either a built in facility of ratio and phase angle correction or necessary interposing Auxiliary current transformers of universal type, shall be provided in the respective panel.

- e) The relay shall be provided with 2nd harmonic restraint or any other inrush proof feature to prevent operation due to magnetizing inrush current when the transformer is charged either from HV or LV side. But this shall not affect the speed of operation for internal fault.
- f) It shall be provided with 5th harmonic restraint features to prevent operation due to possible over excitation of the transformer. This shall also not affect the speed of operation for internal fault.
- g) The relay shall have adjustable bias setting range 20% to 50% and adjustable operating setting range of 10% to 50% at zero bias.
- h) It shall have three instantaneous high set over current units for clearing heavy internal fault.
- i) The relay shall be with 2-bias winding.
- j) The relay shall be such that there will not be any necessity of changing the setting of the relay whenever the transformer taps are changed from +5% to -15%.
- k) Differential relay shall have facility for setting, parameterization, downloading the storage data, data captured by disturbance recorder etc. locally through PC. The necessary Windows based licensed software for establishing the facility to be considered in the scope of the supply by the Bidder.
- l) The relay shall have disturbance recording (with time stamping) function with suitable no. of analog and digital channels, Memory size and number of disturbances stored in the relay shall be clearly indicated in the offer. No. of site selectable BI, BO and watchdog contact details, communication port details (front, rear) along with necessary hardware and software details shall be furnished with the Bid documents.

E-3 RESTRICTED EARTH FAULT PROTECTION

The above protection shall be provided for 33/11 kV transformers at HV and LV side. The Relay shall be: Single pole type.

- a. Current/voltage operated high impedance type with a suitable setting to cover the maximum portion of transformer winding. Necessary calculation to prove the above winding coverage shall be furnished along with the tender.
- b. Tuned to the system frequency.
- c. Have suitable nonlinear resistor to limit the peak voltage and stabilizing resistance.
- d. Operating time shall be less than 40 ms.
- e. Shall be provided in the E/F current element of HV side numerical over current and earth fault rely for HV side. LV side REF protection provided by the differential relay's REF protection element. Both the REF protection scheme shall be high impedance type having suitable stabilizing resistance and metrosil.
- f. Have suitable stabilizing resistor to prevent mal operation during external faults.

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					DE(E), P&CD



E-4 A set of D.C. Voltage Operated Aux. Relays with 4NO self-reset contacts, hand reset having continuous current rating coil for the following transformer internal trouble/fault

- (a) Buchholz Alarm
- (b) Buchholz Trip
- (c) Winding Temp. Trip & winding temp. alarm
- (d) Oil Temp trip & Oil Temp. Alarm
- (e) Low Oil Level Alarm
- (f) Pressure Release Device Trip
- (g) OSR for OLTC trip
- (h) Pressure Release Device of OLTC Trip

E-5 AUXILIARY RELAYS, TRIP RELAYS and TRIP CIRCUIT SUPERVISION RELAYS

Trip Circuit Supervision Relays - All Panels should be provided with D.C. Voltage operated Trip Circuit Supervision Relay having provisions for pre & post close supervision of Trip Circuit with set of self-reset contacts provided for Trip Circuit Healthy Indication and Trip Circuit unhealthy indication & Alarm in respect of Trip Coil/circuits of respective Breakers.

Tripping Relays- All Panels should be provided with D.C. Voltage operated High Speed Tripping Relays having electrical reset contacts capable to make, carry and break trip coil current. Sets of Trip Contacts shall be provided for Inter-tripping function of corresponding 11 kV Incoming Switchgear and closing blocking function of 33 kV & 11 kV Breakers in respect of Transformer Panels. Each set of trip relay shall have minimum two nos. NO and 1No. NC contact as SPARES. The operating time of master trip relay shall be less than 40 ms and electrical reset type with electrical reset contacts and hand-reset flag.

E-6 TRIP CIRCUIT SUPERVISION SCHEME :

Trip circuit supervision scheme shall be such that testing of trip circuit healthiness is possible irrespective of whether the C. B. is in the closed or open position. The Trip Circuit Healthy LED should glow continuously in CB 'ON' Position and in C.B. 'OFF' position. Trip Circuit Supervision relays shall be stand alone type. These relays may be numerical or Static type.

E-7 Principal requirements of protective relays, metering equipments, auxiliary relays etc. are as follows:

E-7-1 Ammeter:

Each circuit one ammeter shall be provided with the following:

Mounting	Flush
Size	96 x 48 mm. case
Response Time	1 second
Operating Temperature	Up to 55°C
Dielectric Strength	2 kV RMS for 1 minute
Auxiliary Supply	230 volt A.C, 50 Hz
Operating Current	1 A from CT Secondary.
Type	Panel Mounting with 3 ¹ / ₂ Digital Display.

E-7-2 Volt Meter :-

Mounting	Flush
Size	96 x 48 mm. Case
Response Time	1 second
Operating Temperature	Up to 55°C
Dielectric Strength	2 kV RMS for 1 minute
Auxiliary Supply	230 V A.C., 50 Hz
Frequency	50 Hz
Operating Voltage	110 V from PT Secondary.
Type	Panel Mounting with 3 ¹ / ₂ Digital Display.

E-7-3 Buzzer

One DC buzzer shall be provided in the BC panel for non-trip alarm. One DC Bell shall be provided in the BC panel for Trip alarm and one AC Bell for Panel DC fail alarm.

E-7-4 High speed tripping relay electrically resettable type confirming to IS - 3231

Aux. voltage	110 V D.C
Coil rating	110V D.C., voltage band for satisfactory operation : 50 to 120% of rated voltage
Operating Time	≤ 40 m. seconds nominal at rated voltage
Burden of relay coil watts (Max)	Low burden 40 Watt at rated voltage
Operating temp	-10 deg C to 55 deg C.
Type and Operational indication	Electrical Reset type and Mechanical red colour Flag Indication
Contact Configuration	6 NO + 4 NC combination Electrical reset contacts and hand-reset flag

Contact ratings:

Make and carry	A.C. 1250 VA with max 5 amp & 660 Volts D.C. 1250 W dc with max 5 amp & 660 Volts
Make and carry for 3 sec.	A.C. 7500 VA with max 30 amp & 660 Volts D.C. 7500 W dc with max 30 amp & 660 Volts
Break	A.C. 1250 VA with max 5 amp & 660 Volts D.C. - 100 W resistive 50 watt inductive with max 5 amp & 660 Volts
Insulation	2 kV RMS, 50Hz for 1 min. 2.5 kV/1 sec between all terminals & case as per IS 3231. 1 kV RMS, 50Hz for 1 min. across open contact
Type of mounting	Flush

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E-7-5 Directional O/C & E/F Relay as per ANNEXURE-A**E-7-6 Numerical based differential protection relay as per ANNEXURE-B****vi. GUARANTEE:**

The panels shall be delivered to the various consignees of the WBSEDCL and shall be suitably packed to avoid damages during transit. The Relays and Meters for each panel are to be delivered to the same consignee. Packing of those items requires adequate care to avoid any possibility of ingress of rain water or seepage of water inside during transportation and subsequent storing at WBSEDCL store/site.

The GIS Panel with relays with all integral parts of the Equipment will be guaranteed for the period of five years from the date of last despatch.

In the event of any defect in the Equipment, relay, any integral part of the Equipment arising out of faulty design, materials, workmanship within the above period, the supplier shall guarantee to replace or repair to the satisfaction of WBSEDCL.

If the supplier fails to do so, within one month of receipt of intimation, WBSEDCL reserves the right to effect repair or replacement by any other agency and recover charges for repair or replacement from the supplier.

38. SCADA Compatibility:

The panels shall be fully SCADA compatible.

39. Type Test:

Manufacturers of HV switchgear shall be able to manage a first party conformity assessment procedure, as defined by ISO 17000, and to provide the associated deliverables "Declaration of Conformity" for the performances stated in this specification. The supplier shall ensure the validity of the declarations over the time. The switchgear must be type tested according to latest relevant IEC standards. The next type tests might be required during tendering process according to the IEC 62271-200 standards.

- a) Tests to verify the insulation level of the equipment.
- b) Tests to prove the temperature rise of any part of the equipment and measurement of the resistance of circuits
- c) Test to prove the capability of the main and Earthing circuits to be subjected to the rated peak and the rated short time withstand currents
- d) Test to prove the making and breaking capacity of the included switching devices.
- e) Test to prove the satisfactory operation of the included switching devices (operation and mechanical endurance test)
- f) Test to verify the protection of persons against access to hazardous parts and the protection of the equipment against solid foreign objects (IP protection degree)
- g) Test to verify the strength of gas filled compartments
- h) Tightness test of gas filled compartments

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40. Routine Tests:

The switchgear shall be subjected to routine tests as per the latest relevant IEC standards. The following tests shall be performed at the supplier's factory:

- a) Visual inspection of: dimensions (according to drawings), painting thickness and adherence, appearance, protection degree, electric devices and polarity of connections.
- b) Electrical tests for low voltage compartment: dielectric test and voltage drop.
- c) Dielectric withstand at power frequency for the entire switchgear.
- d) Voltages drop of main circuit.
- e) Voltages Gas leakage test with helium in every switchgear.
- f) Mechanical/electrical operation and cabling of main circuit and auxiliary circuits.

41. Site Tests:

The complete GIS assembly shall be tested at site after installation, under the manufacturer's supervision. The following site tests shall be carried out:

- a. Visual inspection
- b. Dielectric withstand test at power frequency for the busbar.
- c. Voltage drop test for busbar.
- d. Complete functional tests.

Annexure - D**Standard Make of Relay and Fitments**

1.	Relays	Schneider Infra, ABB, Siemens, Alstom and other reputed relay Manufacturers fulfilling Relay specification as per ANNEXURE-A and ANNEXURE-B and general requirement as per Clause No.22 of the Technical Specification.
2.	Vacuum Interrupter	OEM or CGL / BEL / SCHNEIDER INFRA / ABB / SIEMENS
3.	Breaker Control Switch/ Local-Remote switch	Kaycee/Recom/Switron
4.	Static Ammeter/ voltmeter	AE/RISHAV/Secure
5.	Push Buttons	Vaishno/Teknic/Lumen/STS
6.	Indicating Lamps with lenses	Vaishno/Teknic/Lumen/STS/SWITRON
7.	Panel wiring	Finolex/Havvells/ KEI/ R. R. Kables
8.	Hooter/Buzzer/Bell	Vaishno/STS/JVS/Bharani
9.	Annunciator	MINILEC/ALAN/ INSTALARM/EAPL

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**ANNEXURE-B****Technical specification for IEC 61850 compliant Directional O/C and E/F Relay**

Sl. No.	Feature and Function	Technical requirement
1.	Purpose and application	It is intended to automate the Switchgears specified in the scope of supply and use Communicable Numeric relays for Protection, Control, Metering and Status monitoring. This specification is based on the understanding that an integrated Automation System along with protections shall be provided and same shall have provisions for Integration with SCADA system. All the feeders shall be remote controlled from WBSEDCL's SCADA and from the local console of the numerical relays. Numerical multifunctional combined Microprocessor based Feeder protection and management relay to protect the 33 kV Feeder from all electrical and other faults along with reporting system. Disturbance record for fault analysis. Bidder should comply with any especial requirement or feature asked for retrofitting the relays. Relay should be IEC 61850 compliant. Relay should have 4 CT input, 3 input for O/C and residual E/F protection will be derived internally. One CT input may be used for unbalanced current protection. Relay should have 4 voltage input, 3 input for VT element for directional O/C protection with internally derived residual voltage for E/F protection. Another VT input will be used for residual voltage protection. Relay shall have inbuilt synchro check function. Additional voltage input shall be there to satisfy the Synchro check function. Relay should have two stage over voltage and under voltage protection. The Directional O/C & E/F Relay should have the Site-selectable provision for operating in both Directional & Non-directional mode depending on site requirement.
2.	Main Protection Feature for directional O/C & E/F relay.	1. Electrical over load protection with selectable IEC curves with two stage, first stage to be used as Definite Time / IDMT and second stage to be used as high set for short circuit protection. 2. Earth fault protection in two stages with IEC characteristics. First stage to be used as IDMT/Definite Time and second stage to be used as instantaneous elements. Earth fault element should be suitable for both CBCT and residual type CT connection.

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		3. Negative phase sequence Protection with IEC Curves. 4. CB Fail Protection & time settable as per user. 5. The relay should be immune to DC switching while carrying current i.e. no spurious trip should be generated if relay DC is made On and Off 6. The relay should conform to the IEC255-4 or BS 142 for Inverse time characteristics. 7. VT fuse fail detection on NPS current/NPS Voltage or zero sequence current/voltage based logic and blocking of under voltage protection by VT fuse fail detection. 8. Three phase VT fuses fail detection on current based logic. 9. The relay should have features to monitor for broken conductor and CB opening time. 10. The relay shall be designed for application in WBSEDCL's distribution network where the system is non-effectively earthed through earthing transformer emanating at 33kV bus of 132/33 kV substation. 11. Relay should have minimum two group of setting. Setting group changeover required from digital status input.
3.	Processor feature	Relay shall be completely Numerical with protective elements having software algorithm based on sampling of analog inputs. Sampling Rate of Analog Signal: The sampling rate should be 1000 Hz for 50 Hz signal or better for each analog channel. Hardware based measurements shall not be acceptable.
4.	Operational Philosophy	The operation of Relay shall be possible from both locally from the Switchgear and remote and Local Work station. The local position shall be displayed in remote / local workstation and remote operation shall be blocked if the switch is in Local. Clear control priorities shall prevent initiation of operation of a single switch at the same time from more than one of the various control levels and there shall be interlocks among various control levels. The priority shall always be with the lowest enabled control level. Relay accuracy shall not be affected by system frequency fluctuation.
5.	Status/Optical Inputs/Digital inputs	1. Minimum 30 number status inputs are required

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		- All status inputs should be 110 V DC - Setting group is required to be changed with any input status. - Trip circuit supervision with DI status - The digital inputs shall be acquired by exception with 1ms resolution. Contact bouncing in digital inputs shall not be assumed as change of state. - Relay should have comprehensive self diagnostic feature with remote indication of relay failure and alarm shall be generated without tripping of circuit - Provision of Testing output relays without any current injection. - No. of programmable LEDs- at least 4 nos. with latch option.
6.	Main measuring and reporting feature	All measurements should be in primary quantities. Minimum following displays are required in alpha numeric:- - Three phase (Positive sequence) current, Three phase voltage - Neutral (zero sequence) current, MW, MVAR, Frequency, Pf, MVA etc. - All the trips should have clear indication on the relay terminals - Resetting should be selectable as hand reset or auto reset.
7.	Memory and Recording Feature	- The relay setting and programming should be stored EEPROM so that during Aux. Power failure the said data is not lost. - Relay should have event log, trip log and DR record. All logs should go in to history. - All tripping of relay should initiate DR in auto without extra binary input. Triggering of DR with binary input should be user configurable. - The last 2 fault DR records should be in flash memory and DR

		will not erase in case of DC supply fail for more than 2 days. - Should be able to record at least 5 Oscillographic disturbances and 5 fault records and 250 event records. - Minimum Four no. of latest trip log with cause of trip should be stored in memory along with date and time stamping. The memory should not be lost with the switching off of DC. - The relay should have fault-recording feature with current waveform and Digital Input status. The fault waveform should consist of minimum four current waveforms of three phase current and zero sequence current and DI status. Triggering time for Pre and Post should have user selectable. This record should be in flash memory for minimum 7 days even after switching off the DC supply. - The fault should be date and time stamped. - Communication protocol IEC 61850.
8.	Auxiliary Supply	110 V DC to - 25% to + 10%, 2 wire unearthed system. Necessary software shall be in-built for proper shutdown and restart in case of power failure. Auxiliary supply burden will be around 20Watt.
9.	Rated CT/PT secondary	5/1 Amp(site selectable) , CTs used to be protection class. 3PT input rated 110 Volt (L-L)
10.	Rated frequency	50 HZ +/- 5%
11.	Ambient condition	- Operating ambient temperature up to 55 Deg C - Operating Humidity up to 100 % - Relay shall meet the requirement for withstanding electromagnetic interference according to relevant parts of IEC 61850. Failure of single component within the equipment shall neither cause unwanted operation nor lead to a complete system breakdown.
12.	Module and Mounting	- Relay should be flush mounted type - If module is draw out type then it should have CT shorting facility of make before break type. - Mounting in switchgears located in non AC rooms.

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		4. Galvanic isolation between field connection hardware should be there.
13.	Watchdog and self monitoring	The relay should have facility to monitor the healthiness of circuits and components by own monitoring system. In case of problems, the alarm should be generated by one of the output contacts. The alarm as soft signal to be sent to SCADA system as well. Necessary support documentation explaining the self diagnostic feature shall be furnished.
14.	Settings	Approximate settings possible should be as follows:- 1. Nominal Feeder current 2% to 110 % 2. CT Ratio setting 10-1000(approx.) 3. Earth fault current 5 to 40% with time delay IEC Curves, 2nd stage for instantaneous trip(less than 50 ms) 4. Over current trip- 50% to 200% of 1/5 Amp with time delays as per IEC Curves. 5. High set with delay 200% to 2000% 6. IEC Curves for all O/C and E/F have user selectable. 7. Selectable MTA for Directional features for O/C relay should cover 1st quadrant for effectively grounded system/impedance grounded system/solid grounded system.
15.	Output Relays	Minimum 16 number output relays are required out of which 1. One potential free change over contact should be provided for start inhibit of relay. 2. All o/p contact should be freely programmable. 3. Rating of trip contacts:- a) Contact durability>10K operation b) 15 Amp make and carry for 3 sec for trip contact c) Make and carry for trip contacts L/R<=40ms Rating of Alarm contacts:- d) 8 Amp make and carry continuously for 5 sec.

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		Testing of Output relays through keypad on relay fascia and relay HMI software. Output relay dwell time shall be user programmable or fixed at 100ms.
16.	Relay software and Man Machine Interface	1. The relay should have native IEC 61850 Communication Protocol. 2. Should have password protected key padlock. 3. Necessary software for relay setting, retrieving DR, event log, trip log, and downloading waveform should be supplied by the Bidder. Necessary Licensed is to be issued for WBSEDCL, if required. 4. Bidder has to supply communication hardware for relay setting, DR downloading from front port. This device should be compatible to USB/Ethernet port. 5. It shall be possible to transfer the data stored in the DFR to computer on IEEE/COMTRADE format. The data format shall be compatible for dynamic protection relay testing on relay test kit. COMTRADE Data viewer software is to be provided. 6. Multiuser/Corporate license for installation on minimum 7 nos. of PCs.
17.	Date and time	Date and Time stamping with faults and record. The clock should be powered from internal cell and should not required setting after every DC switching. The internal cell life minimum 5 years. Time synchronization by IRIG-B or SNTP. For time synchronization through SNTP is to be provided from clock signal coming from RTU. In case of IRIG-B, time synchronization will be done with GPS clock signal from GPS receiver located at substation.
18.	Lugs and terminators	All CT and PT terminals shall be provided as fixed (screwed) type terminals on the relay to avoid any hazard due to loose connection leading to CT opening or any other loose connection. Necessary amount of lugs should be supplied along with each relay for CT connection and control wiring. All CT terminals should be capable to connect with ring socket.
19.	Manuals, Drawings and Literature	1. The relays should be supplied with manuals with all technical and operating instructions. 2. All the internal drawings indicating the logics and block

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		diagram details explaining principle of operation given at the time of supply. 3. Mapping details shall be submitted in IEC format.
20.	Standard documentation per Relay, according to IEC 61850	a.i.1. ICS document (model implementation conformance statement) a.i.2. PICS(protocol implementation conformance statement) a.i.3. Conformance Test certificate from KEMA/CPRI. a.i.4. PIXIT document All the above mentioned certificates shall be submitted along with Techno-commercial Bid. a.i.5. ICD file a.i.6. SCD file
21.	Extendibility in Future	The Bidder shall provide all necessary software tools along with source codes to perform addition of bays in future and complete integration with SCADA by the User. These software tools shall be able to configure relay, add analog variable, alarm list, event list, modify interlocking logics etc. for additional bays/equipments which shall be added in future.
22.	Lifespan	The supplier should mention following:- 1. Product maturity: The Bidder should mention the time period for which the product is in the market 2. Expected production life 3. Hardware/Firmware change notification process. Upgrades to be provided free of cost within the Guarantee period, 3 years whichever is later, if needed. 4. Lifespan of standard tools and processes for relay configuration, querying and integration.
23.	Standards	The relay should conform to the IEC255-5 or equivalent BS / ANSI for following:- 1. Overload withstand test 2. Dielectric withstand: 2kV in common, 1 kV in differential

		mode 3. Impulse Voltage: 5kV in common, 1kV in differential mode 4. Insulation resistance > 100 M ohm 5. Vibration: Shock and bump and Seismic 6. Storing and transportation 7. Radio Interference: IEC 61000 for high frequency disturbance, Transient disturbance, Electrostatic discharge 8. KEMA/CPRI Certification for the particular model offered with respect to IEC61850 Protocol
24.	Communication Port	1. Two nos. IEC 61850 (selectable Edition-1 and Edition-2) protocol compliant 100 MBPS Ethernet RJ45/F.O ports having selectable HSR (High Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol) Protocol compliance for communication with SCADA system through two managed Ethernet Switches operating in redundant mode. The communication shall be made in 1+1 mode between individual IED to Switch, such that failure of one set of LAN shall not affect the normal operation of SCADA. However, it shall be alarmed in SCADA. The relay rear ports should accommodate both daisy chain and star bus topology for SCADA integration. 2. Functioning of Relay shall not hamper to fault occurring any interconnected relay. 3. One Front port Ethernet RJ45/USB 2.0 for relay parameterization and configuration etc. with the help of PC. In case RS-232 port offered, suitable interfacing cable with one end having RS 232 port and other end USB 2.0 to be provided to connect with PC free of cost. 4. Relay should generate GOOSE message as per IEC 61850 standard for interlocking and also ensure interoperability with third party relays.
25.	Name Plate and marking	Each IED shall be clearly marked with manufacturer's Name,



		type, serial no. and electrical rating data. Name be made of anodized aluminium with white eng black surface.
26.	Performance Guarantee	Relays will be guaranteed for the period of five years from the of last dispatch. Any problem in the said period should be attended free of charge inclusive of repair/replacement of relays/ component (both H/W, S/W).
27.	Type Test	• Dielectric Withstand Test—IEC 60255-5 • High Voltage Impulse Test, class III --- IEC 60255-5(5kV peak, 1.2/50 micro Sec;3 Positive and 3 negative shots at interval of 5 Sec.) • DC Supply Interruption ---- IEC 60255-11 • AC Ripple on DC supply ---- IEC 60255-11 • Voltage Dips and Short Interruptions --- IEC 61000-4-11 • High frequency Disturbance ---- IEC 60255-22-1, Class III • Fast Transient Disturbance ---- IEC 60255-22-4, Class-IV • Surge withstand capability ---- IEEE/ANSI C 37.90.1(1989) • Degree of Protection • Electromagnetic compatibility • Mechanical stress/vibration test • Temperature withstand Type test reports for the above tests shall be submitted for the approval of WBSEDCL along with Tender. Wherever the above mentioned standards and IEC 61850 overlap, the latter will prevail.
28.	Credential as pre-requisite of Tender	1. Copies of performance certificate for two years successful operation as on the due date of bid opening for the offered relay in respect to implementation of IEC 61850 protocol to any SCADA/substation automation system from reputed Power Sector Utility in India shall have to be furnished along with the Bid. Copies of Purchase Orders and corresponding Delivery Challans /Stores Receipt vouchers/ Excise Duty Invoice, etc., i.e. Proof of Execution of

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	the Purchase Orders. OR Successful testing and operation of minimum one year in WBSEB/WBSEDCL/WBSETCL network. 2. Documentary evidence for being manufacturers like registration Certificate issued by SSI/NSIC/Directorate of Industries/DGS&D, etc. for Qualifying requirement. 3. The bidder should have testing facilities of all functional tests or should have arrangement of all functional tests at government approved testing laboratories. <u>Inter-operability test:-</u> After fulfilment of the above Q.R. inter-operability test of the offered relay (other than Make & Model used in WBSEDCL) with the existing relay in WBSEDCL Network will be tested in WBSEDCL Distribution Testing Department, WBSEDCL for which due intimation for supply of sampled of offered relay will be given to the Bidder. The Bidder needs to submit the said relay to Distribution Testing Department, WBSEDCL within one week from the said intimation. The offered relay will only be accepted after fulfilment of above Q.R. & successful inter-operability test at WBSEDCL system.
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Checklist for Bill of Material for supply

Sl. No.	Material	
1.	Relay (Model No.)	Qty as per Tender
2.	Lugs suitable for current and control, wiring	Qty as per Tender X Number of TBs in relay + 20% extra.
3.	Cable for connection of Relay to laptop(USB port). Along with converter and power supply if required for relay local setting	10 set
4.	Manual, Hard copy in good quality paper properly bounded	10 set

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5.	Copy of Type Test certificate along with manual	With offer
6.	Basic application software for setting change,	10 nos.
7.	CD with software(licensed) to download disturbance recorder, event log and evaluation of those records	10 nos.
8.	Graphical configuration tool for I/P, O/P and functional building block for protection	10 nos.
9.	Any other software required for integration with SCADA.	10 nos.

N.B All the above tools/ Software should be compatible to WINDOWS XP/WINDOWS NT/WINDOWS 7 Operating System.

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ANNEXURE-C**Technical specification for IEC 61850 compliant Differential Relay**

The offered item should meet minimum following requirements:-

Sl. No.	Feature and Function	Technical requirement
1.	Purpose and application	It is intended to automate the Switchgears specified in the scope of supply and use Communicable Numeric relays for Protection, Control, Metering and Status monitoring. This specification is based on the understanding that an integrated Automation System along with protections shall be provided and same shall have provisions for Integration with SCADA system. All the feeders shall be remote controlled from Employer's SCADA and from the local console of the numerical relays. Numerical multifunctional combined Microprocessor based Transformer protection and management relay to protect the 33/11 kV Transformer from all electrical and other faults along with reporting system, Disturbance record for fault analysis. Bidder should comply with any especial requirement or feature asked for retrofitting the relays. Relay should be IEC 61850 compliant. Relay should have 4 current input each for HV and LV side for differential protection of two winding transformer. However in total 7 current inputs may also be acceptable. Relay shall have dedicated current element for high impedance REF protection.

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2.	Main Protection Feature	a.i.1. Relay should have minimum two group of Setting group changeover required from digital status in. a.i.2. Relay should be suitable for differential protection two winding transformers. a.i.3. Relay should have 2nd and 5th harmonic blocking feature and unrestrained High Set current tripping feature. a.i.4. Relay may have backup O/C and E/F feature with IDMTL/Definite Time. There should be option for derivation of E/F internally. a.i.5. CB Fail Protection & time settable as per user. a.i.6. The relay should be immune to DC switching while carrying current i.e. no spurious trip should be generated if relay DC is made ON and OFF. a.i.7. The relay should conform to the IEC255-4 or BS 142 for Inverse time characteristics.
3.	Processor feature	Relay shall be completely Numerical with protective elements having software algorithm based on sampling of analog inputs. Sampling Rate of Analog Signal: The sampling rate should be 1000 Hz for 50 Hz signal. Hardware based measurements shall not be acceptable.
4.	Operational Philosophy	The operation of Relay shall be possible from both locally from the Switch gear and remote and Local Work station. The local position shall be displayed in remote / local workstation and remote operation shall be blocked if the switch is in Local. Clear control priorities shall prevent initiation of operation of a single switch at the same time from more than one of the various control levels and there shall be interlocks among various control levels. The priority shall always be with the lowest enabled control level.
5.	Status/Optical Inputs/Digital inputs	1. Minimum 16 numbers status inputs are required 2. All status inputs should be 110 V DC 3. Setting group is required to be changed with any Digital input status.

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		4. Trip circuit supervision with DI status 5. The digital inputs shall be acquired by exception with 1ms resolution. Contact bouncing in digital inputs shall not be assumed as change of state. 6. Relay should have comprehensive self diagnostic feature with remote indication of relay failure and alarm shall be generated without tripping of circuit 7. Provision of Testing output relays without any current injection. 8. No. of programmable LEDs- at least 4 nos. with site - selectable latching option.
6.	Main measuring and reporting feature	All measurements should be in primary quantities. Minimum following displays are required in alpha numeric:- 1. Three phase (Positive sequence) current and Phase angle (Both HV and LV side) 2. Neutral(zero sequence) current, 3. All the trips should have clear indication on the relay terminals. 4. Resetting should be selectable as hand reset or auto reset.
7.	Memory and Recording Feature	1. The relay setting and programming should be stored in EEPROM so that during Aux. Power failure the said data is not lost. 2. Relay should have event log, trip log and DR record. All logs should go in to history. 3. All tripping of relay should initiate DR in auto without extra binary input. Triggering of DR with binary input should be user configurable. 4. The last 2 fault DR records should be in flash memory and DR will not erase in case of DC supply fail for more than 2 days. 5. Minimum Four no. of latest trip log with cause of trip should be stored in memory alone with date and time stamping. The memory should not be lost with the switching off of DC.

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		6. The relay should have fault-recording feature with waveform and Digital Input status. The fault waveform should consist of minimum four current waveforms of phase current and zero sequence current and DI status. Triggering time for Pre and Post should have user selectable. This record should be in flash memory for minimum 7 days even after switching off the DC supply. 7. The fault should be date and time stamped. 8. Communication protocol IEC 61850.
8.	Auxiliary Supply	110 V DC with variation of - 25% to + 10%, 2 wire unearthed system
9.	Rated CT secondary	5/1 Amp (site selectable), CTs used to be protection class.
10.	Rated frequency	50 HZ +/- 5%
11.	Ambient condition	1. Operating ambient temperature up to 55 Deg C 2. Operating Humidity up to 100 % 3. Relay shall meet the requirement for withstanding electromagnetic interference according to relevant parts of IEC 61850. Failure of single component within the equipment shall neither cause unwanted operation nor lead to a complete system breakdown.
12.	Module and Mounting	1. Relay should be flush mounted type 2. If module is draw out type then it should have CT shorting facility of make before break type. 3. Mounting in switchgears located in non AC rooms.
13.	Watchdog and self monitoring	The relay should have facility to monitor the healthiness of its circuits and components by own monitoring system. In case of any problems, the alarm should be generated by one of the output contacts. The alarm as soft signal to be sent to SCADA system as well. Necessary support documentation explaining the self diagnostic feature shall be furnished.
14.	Settings	Approximate settings possible should be as follows:- 1. Nominal Feeder current 2% to 110 % 2. CT Ratio setting 10-1000(approx.)

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		3. Relay should have amplitude matching features by means of reference power common to all winding. 4. Relay should have site selectable suitable for all available vector groups. It will automatically taken care of zero sequence filtering of relay currents depending on relevant chosen vector group. 5. The bias operating characteristics of the relay shall have at least two knees and site selectable highest tripping facility (8 to 15 times of Relay basic current). Relay should have site selectable percentage bias setting from 20% to 45%. The differential operating current of the relay shall have site selectable range from 10% to 40%. 6. Relay shall have through fault stabilization features up to at least 10 times of relay basic current in the event of through fault. 7. Relay should have 2nd harmonic restraint features in order to bring stabilization under inrush condition. The percentage 2nd harmonic restraint features shall have site selectable setting range from 15% to 40%. 8. The relay shall have over fluxing restraint features based on site selectable ratio of 5th harmonic to fundamental current. The relay shall have also facility to activate/ deactivate the above features in site.
15.	Output Relays	Minimum 16 number output relays are required out of which 1. One potential free change over contact should be provided for start inhibit of relay. 2. All o/p contact should be freely programmable. 3. Rating of trip contacts:- iii.a) Contact durability > 10K operation iii.b) 15 Amp make and carry for 3 sec for trip contact iii.c) Make and carry for trip contacts $L/R \leq 40\text{ms}$ Rating of Alarm contacts:- a) 5 Amp make and carry continuously for 5 sec.

16.	Relay software and Man Machine Interface	1. The relay should have native IEC 61850 Comm Protocol. 2. Should have password protected key padlock. 3. Necessary software for relay setting, retrieving DR, event log, trip log, downloading waveform should be supplied by the Bidder. Necessary Licensed is to be issued for WBSEDCL, if required. 4. Bidder has to supply communication hardware for relay setting, DR downloading from front port. This device should be compatible to USB/Ethernet port. 5. It shall be possible to transfer the data stored in the DFR to computer on IEEE/COMTRADE format. The data format shall be compatible for dynamic protection relay testing on relay test kit. 6. Should be able to record at least 5 Oscillographic disturbances and 5 fault records and 250 event records.
17.	Date and time	Date and Time stamping with faults and record. The clock should be powered from internal cell and should not required setting after every DC switching. The internal cell life minimum 5 years. Time synchronization by IRIG-B or SNTP.
18.	Lugs and terminators	All CT terminals shall be provided as fixed (screwed) type terminals on the relay to avoid any other any hazard due to loose connection leading to CT opening or any other loose connection. Necessary amount of lugs should be supplied along with each relay for CT connection and control wiring. All CT terminals should be capable to connect with ring socket.
19.	Manuals, Drawings and Literature	1. The relays should be supplied with manuals with all technical and operating instructions. 2. All the internal drawings indicating the logics and block diagram details explaining principle of operation should be given at the time of supply. 3. Mapping details of all the details shall be submitted in IEC format.
20.	Standard documentation per Relay, according to IEC	1. MICS document (model implementation conformance statement)

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	61850	- PICS(protocol implementation conformance statement) - Conformance Test certificate by KEMA/CPRI All the above mentioned Certificates shall be submitted along with the Techno-commercial Bid - ICD file - SCD file
21.	Extendibility in Future	The Bidder shall provide all necessary software tools along with source codes to perform addition of bays in future and complete integration with SCADA by the User. These software tools shall be able to configure relay, add analog variable, alarm list, event list, modify interlocking logics etc. for additional bays/equipment which shall be added in future.
22.	Lifespan	The supplier should mention following:- - Product maturity: The Bidder should mention the time period for which the product is in the market - Expected production life - Hardware/Firmware change notification process. Upgrades to be provided free of cost within the warranty period/5 years, if needed. - Lifespan of standard tools and processes for relay configuration, querying and integration
23.	Standards	The relay should conform to the IEC255-5 or equivalent BS / ANSI for following:- - Overload withstand test - Dielectric withstand: 2kV in common, 1 kV in differential mode - Impulse Voltage: 5kV in common, 1kV in differential mode - Insulation resistance>100 M-ohm - Vibration: Shock and bump and Seismic - Storing and transportation - Radio Interference: IEC 61000 for high frequency

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		disturbance, Transient disturbance, Electrostatic discharge
		8. KEMA Certification for the particular model offered in respect to IEC61850 Protocol.
24.	Communication Port	1. Two nos. IEC 61850 (selectable Edition-1 and Edition-2) protocol compliant 100 MBPS Ethernet RJ45/F.O ports having selectable HSR (High Availability Seamless Redundancy) and PRP (Parallel Redundancy Protocol) Protocol compliance for communication with SCADA system through two managed Ethernet Switches operating in redundant mode. The communication shall be made in 1+1 mode between individual IED to Switch, such that failure of one set of LAN shall not affect the normal operation of SCADA. However, it shall be alarmed in SCADA. The relay rear ports should accommodate both daisy chain and star bus topology for SCADA integration. 2. Functioning of Relay shall not hamper to fault occurring any interconnected relay. 3. One Front port Ethernet RJ45/USB 2.0 for relay parameterization and configuration etc. with the help of PC. 4. Relay should generate GOOSE message as per IEC 61850 standard for interlocking and also ensure interoperability with third party relays.
25.	Name Plate and marking	Each IED shall be clearly marked with manufacturer's Name, type, serial no. and electrical rating data. Name plates shall be made of anodized aluminium with white engraving on black surface.
26.	Performance Guarantee	Relays will be guaranteed for the period of five years from the date of last dispatch. Any problem in the said period should be attended free of charge inclusive of repair/replacement of relays/ component (both H/W, S/W).
27.	Type Test	As per T/S of Directional O/C & E/F Relay Specification
28.	Credential as pre-requisite of Tender	1. Copies of performance certificate for two years successful operation as on the due date of bid opening for the offered relay in respect to implementation of IEC 61850 protocol to any SCADA/substation automation system from reputed Power Sector Utility in India shall have to be furnished along with the

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		Bid. Copies of Purchase Orders and corresponding Delivery Challans /Stores Receipt vouchers/ Excise Duty Invoice, etc., i.e. Proof of Execution of the Purchase Orders. OR Successful testing and operation of minimum one year in WBSEB/WBSEDCL/WBSETCL network.. 2. Documentary evidence for being manufacturers like registration Certificate issued by SSI/NSIC/Directorate of Industries/DGS&D, etc. for Qualifying requirement. 3. The bidder should have testing facilities of all functional tests or should have arrangement of all functional tests at government approved testing laboratories. <u>Inter-operability test:-</u> After fulfilment of the above Q.R. inter-operability test of the offered relay (other than Make & Model used in WBSEDCL) with the existing relay in WBSEDCL Network will be tested in WBSEDCL Distribution Testing Department, WBSEDCL for which due intimation for supply of sampled of offered relay will be given to the Bidder. The Bidder needs to submit the said relay to Distribution Testing Department, WBSEDCL within one week from the said intimation. The offered relay will only be accepted after fulfilment of above Q.R. & successful inter-operability test at WBSEDCL system.
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Checklist for Bill of Material for supply

Sl. No.	Material	
1.	Relay (Model No.)	Qty as per Tender
2.	Lugs suitable for current and control, wiring	Qty as per Tender X Number of TBs in relay + 20% extra.
3.	Cable for connection of Relay to laptop (USB port). Along with converter and power supply if required	10 set

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	for relay local setting	
4.	Manual, Hard copy in good quality paper properly bounded	10 set
5.	Copy of Type Test certificate along with manual	With offer
6.	Basic application software for setting change,	10 nos.
7.	CD with software(licensed) to download disturbance recorder, event logger collection and evaluation of those records	10 nos.
8.	Graphical configuration tool for I/P, O/P and functional building block for protection	10 nos.
9.	Any other software, if required for integration with SCADA	10 nos.

N.B All the above tools/ Software should be compatible to WINDOWS XP/WINDOWS NT/WINDOWS 7 and other WINDOWS latest version Operating System.

SCHEDULE - VII

Undertaking from Relay manufacturer

We hereby confirm that the main protective relay(s) type _____

i)

ii)

iii)

offered by us against your tender No. _____ through M/s. _____ are in our current range of production. We also confirm that these relays will not be phased out by us in the next 10 years from the date of supply. Necessary repairs/replacements if necessary during this period will be made available by us.

Name & Designation: _____

Company Seal: _____

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42. GUARANTEED TECHNICAL PARTICULARS FOR SWITCHGEAR (33 KV GIS):

Sl.No.	Particulars	33 kVGIS (Cubicle type)
1.a	Type (Model No.)	To be specified by the bidder.
1.b	Standard Applicable	IEC-62271-100/ IEC-62271-200
2.	Service	Indoor
3.a	Enclosure-Tank	Stainless Steel/high grade Aluminum Alloy
3.b	Enclosure- Panel	CRCA/Alu-zink
4.	Nominal System Voltage	33 kV
5.	Highest System Voltage	36 kV
6.	No. of phases and frequency	3ph.50Hz
7.	Busbar material	Copper
8.	Bus Color code	RYB
9.	System Earthing	Impedence earthed
10.	Circuit Breaker Rating	1250 A
10.1	Continuous Current Rating at 50°C	1250 A for CB & Isolator & 1600 A for busbar
10.2	Short Circuit Rating	25 kA
10.3	Short Circuit duration	3 sec
10.4	Internal Arc Rating (Internal Arc Classification of IAC-A-FLR)	25 kA
10.5	Internal Arc Duration	1 sec
11.	Rated making Current	As per IEC-62271
12	Operating duty	O-0.3sec-CO-3minutes-CO
13	Leakage rate per year in gas compartment	Less than 0.1%
14.	Busbar Current rating	1600 A

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15.	DS rating	1250 A
16.	Power Frequency Withstand voltage	70 kV for 1 minute
17.	Impulse withstand voltage(1.2/50 microsec)	170 kV
18.	Control Voltage	110 V DC
19.	Spring charge motor voltage	110 V DC
20.	CT Particulars	
	A) All feeders panels	
	i) Ratio	400-200 / 1-1 A
	ii)Core- 1 : Metering	VA: 10 ; CL: 0.5; ISF<=5 at LR
	iii)Core-2: Protection	VA: 10 ; CL: 5P; ALF: 15
	B) Transformer panels	
	i) Ratio	400-200 / 1-1-1 A
	ii)Core- 1 : Metering	VA: 10 ; CL: 0.5; ISF<=5 at LR
	iii)Core-2: Protection	VA: 10 ; CL: 5P; ALF: 15
	i) Core-3: PS	VA:10; CL: PS; V _k > 70 V at LR; I _{mag} ,<=30mA at V _k /2 at LR
21.	PT ratio-STAR/ STAR	$(33/\sqrt{3})/ (.11/\sqrt{3})/ (.11/\sqrt{3})/.11/3$
22.	Aux. Contacts	6 NO + 6 NC
23.	Termination	Plug in type/push on, according to DIN 47673 outer cone/inner cone type as per relevant IEC. Provision shall be made to connect at least two nos. of specified cable size may connect in each bay.
23.1	Incomers	XLPE Cables as specified
23.2	Outgoings	XLPE Cables as specified
24.	Degree of protection (HV equipment)	IP-65 for GasCompartment
25	Maximum Dimensions (mm)	1785 (D) X 800 (W) X 3000 (H)

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43. SCHEDULE OF GUARANTEED TECHNICAL PARTICULARS OF 33KV GIS FOR BREAKERS/ PANELS:

01.	Manufacturer's Name and Country of origin	
01.1	Manufacturing Facilities for GIS in INDIA	
01.2	Manufacturing Location	
02.	Manufacturer's Design/type Ref	
03.	Frequency	
04.	Rated Voltage	
05.	Highest system voltage	
06.	Rated current	
07.	Short Circuit current rating with duration	
08.	Certificate or report of short circuit type test	
09.	Rated operating duty cycle	
10.	Short Circuit Breaking Current: (a) Symmetrical cal (b) Symmetrical at rated voltage (c) Asymmetrical at rated voltage (i) Per Phase (ii) (ii) Average (iii) D.C. Component	
11	Arcing time (at rated breaking current) in ms.	
12	Opening time	
13	Total break time in millisec.	

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	(a) At 10% rated interrupting capacity (b) At rated interrupting capacity	
14.	Make time in ms.	
15.	Dry 1 minute power frequency withstand test voltage(a)Between line terminal and Earth KV rms (b)Between terminals with breaker contacts open	
16.	1.2/50 full wave impulse withstand test voltage (a) Between line terminal and Earth KV rms (b)Between terminals with breaker contacts open Kvp	
17	Control Circuit Voltage DC	
18	Power required for Closing Coil at 110 V	
19	Power required for Tripping Coil at 110 V	
20	Whether Trip free or not	
21	Whether all the interlocks provided	
22	Overall dimensions	
23	Total weight of one complete Breaker	

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44. SCHEDULE OF GUARANTEED TECHNICAL PARTICULARS FOR 33KV CURRENT TRANSFORMERS:

01.	Manufacturer's Name and country of origin	
02.	Manufacturer's design Ref/ Model	
03.	Applicable Standards	
04.	Type	
05.	Rated Primary current	
06.	Rated secondary current	
07.	Rated frequency	
08.	Transformation ratios	
09.	Number of cores	
10.	Rated output (Core wise)	
11.	Class of insulation	
12.	Class of accuracy (a)For metering (b)For Protection (c) PS Class	
13.	Short circuit current rating and its duration	
14.	One minute power frequency dry withstand voltage	
15.	1.2/50 microsec. Impulse withstand test voltage	
16.	One minute power frequency withstand test voltage on secondary	
17.	Instrument safety factor	
18.	Type of primary winding	

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45. SCHEDULE OF GUARANTEED TECHNICAL PARTICULARS FOR 33kV
POTENTIAL TRANSFORMERS:

01.	Manufacturer's Name and country of origin	
02.	Manufacturer's design reference	
03.	Applicable Standards	
04.	Type	
05.	Ratio	
06.	Rated Primary voltage	
07.	Rated secondary voltage	
08.	Rated frequency	
09.	Class of accuracy	
10.	No. of phase and method of connection	
11.	Burden	
12.	One min. power frequency dry flash over voltage	
13.	1.2/50 micro sec impulse withstand test voltage	
14.	Class of insulation	

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46. SCHEDULE OF GUARANTEED TECHNICAL PARTICULARS FOR NUMERICAL RELAYS:

01.	Manufacturer's Name and country of origin		
02.	Manufacturer's design Ref/Type		
03.	Applicable Standards		
04.	Current setting range for		
	(a) Over current relay	IDMTL Instantaneous	
	(b) Earth-fault relay	IDMTL Instantaneous	
	(c) Contact Rating		
05.	Details on IDMTL characteristics		
06.	Whether High Set is Transient free		
07.	Whether separate Time setting for IDMTL/Instantaneous Elements available		
08.	Whether Relay senses True RMS Current		
09.	Accuracy for different settings and limits of errors		
10.	Whether settings site selectable and HMI provided		
11.	Whether Alpha Numeric LED display		
12.	Whether Compatible for 110 VDC		
13.	Whether Compatible for 1A CT Secondary		
14.	Whether Self diagnostic features available		
15.	Nos. and types of Communication PORT available		
16.	Whether Blocking characteristics available for blocking the unscrupulous tripping of Upstream Breakers		

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17.	a) Whether relay test block is provided b) Type of test block with literature	
18.	Whether draw out type unit or not	
19.	Types of case	
20.	Reset time	
21.	Burden of relay	
22.	Maximum and Minimum operating ambient air Temperature	

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